The Pioneer School Program: Education Reform in Morocco *

Ryan McWay†
Paul Glewwe‡
Mridul Joshi§
Youssef Assarssah¶
Kenza Sabri||

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To address the global learning crisis, targeted remediation and structured pedagogy are often considered "great buys" for education policymakers. But successful small-scale programs can be difficult to scale, so that impacts at scale in government schools are often disappointing. We collaborated with Morocco's Ministry of Education to conduct a prospective evaluation of Morocco's Pioneer School Program (PSP), a large-scale education reform that combines two "great buys". We implement a matched-paired difference in differences design on 276 schools that implemented the PSP in 2023 (Cohort 1 schools) and 62 schools that implemented the PSP in 2024 (Cohort 2 schools). We follow 184 of the Cohort 1 schools into the second year of the reform, presenting results on the effects of the PSP over two years in these schools. We also estimate the impact after one year for the 62 Cohort 2 schools. For the Cohort 1 schools after two years, the program resulted in large increases in student learning of Arabic (0.34 s.d.), French (0.91 s.d.), and mathematics (0.59 s.d.). The overall impact of 0.59 s.d. across all subjects, as well as the overall average impact of 0.46 s.d. for the second cohort after 1 year of exposure are smaller than the impacts in the Year 1 report. Although, the impact of the PSP still remains very large relative to impacts that have been found for education programs in other middle- and low-income countries.

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†PhD student, Department of Applied Economics, University of Minnesota. E-mail: mcway005@umn.edu

‡Distinguished McKnight Professor, Department of Applied Economics, University of Minnesota. E-mail: pglewwe@umn.edu.

§PhD candidate, Graduate School of Education, Stanford University . E-mail: mriduljoshi@stanford.edu

¶Research fellow, Harvard Kennedy School, Harvard University. E-mail: youssef_assarssah@hks.harvard.edu

||Research associate, Morocco Innovation and Evaluation Lab, J-PAL Middle East and North Africa. E-mail: ksabri@povertyactionlab.org

1 Introduction

More than half of children in low- and middle-income countries do not learn to read with comprehension by age 10 (World Bank, 2019). Similarly, poor performance has been found in students' basic mathematics skills (Angrist et al., 2021). The World Bank (2018) describes this state of affairs as a global "learning crisis". While nearly all of these countries have enrolled almost all of their children in primary school, progress in terms of student learning in these schools is often disappointing. What can be done to increase student learning in these countries? Recent research has provided evidence on which educational interventions increase student learning in low- and middle-income countries. The Global Education Evidence Advisory Panel (GEEAP, 2023) conducted a systematic review of the evidence to identify "great buy" interventions, which are defined as interventions that have been found to be highly cost-effective based on a large body of evidence. Unfortunately, many promising interventions have been found to be less effective when program implementation is transferred from non-governmental organizations to governments (Vivalt, 2020; Allcott, 2015). This observation holds for two of the above-mentioned "great buy" recommendations: scripted lesson plans (Kerwin and Thornton, 2021; Piper and Dubeck, 2024) and targeting instruction to a child's learning level (Banerjee et al., 2017; Duflo et al., 2024).

This study estimates the impact of Morocco's Pioneer School Program (PSP) on student learning after two years. The PSP combines two of the GEEAP "great buy" recommendations for increasing student learning in low- and middle-income countries: structured pedagogy (including scripted lessons), and targeted remediation (by student learning level, not grade). In addition, the PSP assigns some primary school teachers to specialize in teaching Arabic, French, and mathematics, and it recognizes schools that successfully participate in the program (with a system of quality certification).[1] The program was launched in 626 of Morocco's public primary schools serving 320,000 primary students in September 2023. One year later it expanded to another 2,000 schools, so that is was operating in 2,626 schools in September 2024 serving 1.3 million students. This study evaluates the program's impact on student learning for these two sets of schools over the first two years of the program's implementation. This paper complements the study by Ibrahim et al. (2025), which found, after only one year, very large impacts of the PSP (0.9 s.d. averaged across subjects) on students' Arabic, French and mathematics skills for a subset of the 626 schools that started the PSP in September of 2023.

When evaluating any program, credible estimates of the program's impact require a valid counterfactual (viz., comparison group). Our study relies on a difference-in-differences strategy. Using administrative data on all public primary schools in Morocco, Ibrahim et al. (2025) applied machine learning methods to select a similar non-PSP primary school for each of the 626 PSP schools that implemented the PSP in September of 2023. From these 626 pairs of schools, 138 of the closest matched pairs were selected for the study of Ibrahim et al. (2025). We call these the Cohort 1 schools. In this study we follow 92 pairs of these Cohort 1

[1]The criteria for school recognition in the PSP is a ministerial commission label awarded as 'the École Pionnière' (Pioneer School). Certification is based on the ONDH (National Observatory of Human Development) evaluation reports covering three pillars: (1) Students: TaRL implementation and the remediation period, (2) Teachers: Specialization, training, certification, and one full year of implementation monitored by inspectors, (3) Schools: Infrastructure and broader program-level requirements.

schools for a second year (46 pairs were dropped because the comparison school in those pairs implemented the PSP program in September of 2024). We then supplemented these 92 pairs with 31 pairs of schools which include one school that started the PSP in September of 2024 and a similar comparison school that was matched to the treatment school using the same machine learning methods that were used by Ibrahim et al. (2025). We call these schools the Cohort 2 schools. Initially, we selected 70 potential schools (35 PSP schools and 35 comparison schools) to survey. At baseline, we surveyed only 67 of these schools; three were not surveyed due to logistical issues. The corresponding matched pairs for these schools were also dropped, limiting our sample to 64 Cohort 2 schools in our evaluation after one year of PSP participation. Finally, we drop another pair of schools as one of the PSP schools was found not to have properly implemented the PSP program, invalidating its treatment status. For both the second year of Cohort 1 and the first year of Cohort 2, we collected information for grades 2 through 6. To estimate the PSP program's impact in its second year, we collected primary assessment data and compared changes in student learning over time in the PSP schools to the same changes for the matched non-PSP schools, separately for the two cohorts of schools. For both cohorts, the key assumption is that the average (conditional) trend in the PSP schools would have been the same as the average (conditional) trend in the matched non-PSP schools had the PSP schools not participated in the program. For Cohort 1 schools, comparisons of trends in exam scores over the seven years before the program was implemented support this "parallel trends" assumption. For the Cohort 2 schools, similar comparisons for the same seven years before the start of the PSP program also support this assumption.

In the first year endline report (Ibrahim et al., 2025), estimates for the Cohort 1 schools yielded very large intent-to-treat (ITT) impacts of the PSP program on student learning. We replicate the year 1 results in our study for Cohort 1. Averaging over all three subjects (Arabic, French and mathematics), the impact of the PSP on student learning is 0.90 standard deviations (s.d.) of the distribution of the scores of the non-PSP schools at endline. By subject, the program's impacts are 0.52 s.d. for Arabic, 1.30 s.d. for French, and 0.93 s.d. for mathematics. The estimated impacts are very similar for female and male students, with slightly larger impacts for female students in French and mathematics, and for the overall score. Disaggregating by baseline student performance (within each grade level), the impacts on Arabic are higher for students in the bottom 25 percent of students in a given grade's baseline distribution (0.63 s.d.) than for students in the top 25 percent of that distribution (0.41 s.d.), while the impacts on mathematics are lower (but still extremely high) for students in the bottom 25 percent (0.85 s.d.) than for students in the top 25 percent (0.98 s.d.). For French, the impacts are slightly higher for students in the bottom 25 percent of students in that distribution (1.23 s.d.) than for students in the top 25 percent (1.15 s.d.).

In this report, we follow a subset of these Cohort 1 schools into the second year of exposure to the PSP program. After two years, averaging over all three subjects, the impact of the PSP program is 0.59 s.d. of the distribution of scores of the non-PSP schools at (year 2) endline. While this impact is smaller than the impact after one year found by Ibrahim et al. (2025), it is still a very large learning impact over two years. By subject, the program's impact after two years is 0.34 s.d. in Arabic, 0.91 s.d. in French, and 0.59 s.d. in mathematics. Again, the impact across gender is very similar with female students benefiting slightly more (0.63 s.d.) than male students (0.55 s.d.). Considering baseline performance, impacts for Arabic are

higher for students in the bottom 25 percent of students given grade's baseline distribution (0.35 s.d.) than for students in the top 25 percent of that distribution (0.26 s.d.), while the impact on French is lower for students in the bottom 25 percent (0.74 s.d.) than for students in the top 25 percent (0.88 s.d.). The impact for mathematics was very similar across the distribution of baseline student performance.

To assess the effect of the expansion of the PSP in September 2024, we estimate the impact of one year of exposure to the PSP on the students in the Cohort 2 schools. The average impact of the PSP over all three subjects is 0.46 s.d.; this is only about half of the effect size of the estimated impact on students in Cohort 1 schools after one year. This remains very large when compared to impacts of education programs that have been estimated for other developing countries as exhibited by Figure 2. By subject, the PSP impact is 0.28 s.d. for Arabic, 0.65 s.d. for French, and 0.49 s.d. for mathematics. To better understand the initial effect of the PSP as the program is rolled out, we estimated a pooled effect of the first years exposure across Cohorts 1 and 2. The combined effect across the two cohorts for the first year averaged across the three subjects is 0.68 s.d. This indicates that the PSP program still yields one of the largest impacts of any program on student learning in mathematics and reading (Evans and Yuan, 2022).

We conduct an exploratory analysis on the mechanisms which may be driving the reduction in the impact of the program from year 1 to year 2. While still outperforming comparison school students in both years, the PSP students did not continue to increase their test performance in year 2. In particular, they performed about the same at endline in the second year as they did at endline in the first year. In contrast, the comparison school students experienced a modest improvement from year 1 to year 2 in their endline test performance. This provides a clearer picture of the reduction in the test score gap between the comparison group students and the PSP students from year 1 to year 2.

We find evidence for a potential reason for the decline in the program's impact over a two year exposure by focusing primarily on Cohort 1. We show there is some minor evidence that the PSP test scores suffer from ceiling effects. The top performing students were reaching the highest possible scores (ceiling) in year 1, so there was no room for further improvement year 2. So the diminished gap between PSP and comparison students from year 1 to year 2 is partly explained by the ceiling that the top performing PSP students reached in year 1, which prevented them from increasing their scores further in year 2. This is only evident for French test performance; not Arabic or mathematics test performance. Even so, there remained plenty of room to increase test scores after year 1 for most of the students in the PSP schools, so this ceiling effect cannot be the only explanation for why the test score gap between PSP students and comparison group students decreased from year 1 to year 2.

2 The Pioneer School Program (PSP)

Public provision of education, as governed by the Ministry of Education (MENPS or Ministère de l'Education Nationale, du Préscolaire et des Sports), is the most common type of primary school education in Morocco. While the share of private enrollment has increased

in recent years, as of 2022, public schools still enrolled 83 percent of primary school students in Morocco. Primary school enrollment numbers are high; the net enrollment rate is 99 percent. Persistence is also high; as of 2021, 98 percent of enrolled primary school students persisted to grade 6, the last grade of primary school (UIS-UNESCO, 2024).

However, two major concerns for education in Morocco are low continuation rates into secondary school and low student performance in both primary and secondary school. Only about 70 percent of Moroccan students who start primary school transition from primary to secondary education (Mansouri and Moumine, 2017). Equally – if not more – concerning are low levels of student learning. In the 2021 PIRLS reading assessment, despite recent progress, Morocco's fourth-graders ranked second to last (out of 57 countries), and more than half of the students (59 percent) did not even reach the minimum proficiency benchmark (PIRLS, 2021). Similarly, in the 2019 TIMSS mathematics assessment of fourth grade students, Morocco ranked among the lowest-performing countries (60 out of 64), and more than half of the students (57 percent) did not even reach the minimum proficiency benchmark (Mullis et al., 2019).

The Pioneer School Program (PSP) was inspired by the report of the Global Education Evidence Advisor Panel (GEEAP, 2023). The PSP combines two of that report's "great buy" recommendations on how to increase student learning in low- and middle-income countries. The PSP was launched in 626 of Morocco's public primary schools in September 2023, and it expanded to another 2,000 primary schools in September 2024.[2] The Ministry of Education considers the program to be its flagship intervention for primary schools, and the remaining primary schools are expected to be reached over the following two academic years (2025/2026 and 2026/2027), for a four-year roll-out of the education reform.

The program's first component is targeted remediation for students in grades 2 through 6, following the Teaching at the Right Level (TaRL) approach.[3] The Ministry of Education implemented this remedial component in PSP schools during a dedicated, two-month period at the beginning of the 2023 – 24 school year for the first cohort of schools, and a six-week period at the start of the 2024 – 25 school year for the second cohort of schools. Thereafter, the program included weekly follow-up activities throughout the school year.[4]

The second program component consists of training teachers on how to implement structured pedagogy in their regular classes in grades 1 to 6. To do this, the program provided teachers with scripted lesson plans, projectors, and a sequence of readily prepared classes for teachers to deliver through slide decks they receive from the Ministry.

The program compliments these first two components by allowing schools to assign some of their primary school teachers to specialize in teaching Arabic, French, and mathematics

[2]These schools had volunteered to participate in the program, and a slightly smaller subset of these schools were then selected by the Ministry. Nearly all schools that volunteered were accepted into the program by meeting the adequate conditions to join (e.g., sufficient building conditions).

[3]The Ministry of Education refers to its program as TaRL, but TaRL Africa and Pratham (the organizations that promote the Teaching at the Right Level program in India and multiple African countries) are not participating in the program's implementation.

[4]Regarding the two-month period in the 2023-24 school year, the original plan was to have a six-week period, but it increased to two months due to a teachers strike. Also, while implementation of weekly follow up activities was intended, there is qualitative evidence that there was little supervision by the Ministry of Education, and so relatively few schools implemented the meetings continuously throughout the academic year.

and by recognizing schools that successfully participate in the program through a system of quality certification. These are the third and fourth components of the PSP. The schools that obtain a "Pioneer School" label receive an annual allowance in the first year of PSP participation. All of the program implementers are regular teachers and inspectors commonly assigned to the program schools, with no additional staff being assigned to schools and no NGO involvement in the program.

3 Methodology

This report on the impact of the PSP after two years presents estimates of the intent-to-treat effect of the PSP intervention on student learning and several other education outcomes using a strategy that combines difference-in-differences estimation with matching methods. This strategy compares changes over time in the outcome variables of interest for primary schools that were assigned to receive that intervention with the same changes of their matched comparison schools. For all outcomes, the following empirical specification is used:

$$Y_{igsjt} = \beta(TREAT_{sj} \times POST_t) + \delta(X_{igsj} \times POST_t) + \zeta(\eta_j \times \theta_g \times POST_t) + \gamma_{igsj} + \varepsilon_{igsjt} \quad (1)$$

Here, Y_{igsjt} is outcome Y for student i in grade g in school s in matched pair j in period t. $TREAT_{sj}$ is the assignment of school s to receive the education intervention, and $POST_t$ is a dummy variable equal to 1 if the outcome Y is measured at endline (i.e., after the PSP was rolled out) and 0 if measured at baseline (i.e., before the PSP was rolled out).

The $\gamma_{i,g,s,j}$ term is a student fixed effect that measures the impacts on Y_{igsjt} of all student characteristics, observed or unobserved, that do not change over time. However, the impacts of some student characteristics may change over time, so a vector of observed student characteristics, denoted by X_{igsj}, is added and multiplied by $POST_t$. The corresponding vector of coefficients for this interaction term, denoted by δ, measures the change in the impacts of these observed student characteristics over time. In other words, δ measures the contributions of observed student characteristics to changes in the outcome variables over time. Since nearly all students stay in the same school between baseline and endline data collection, student fixed effects also include school fixed effects, and so the X_{igsj} variables can also include observed school variables. The student variables available in the dataset are students' gender and whether the student was eligible for a social safety net program (measured at baseline). The school characteristics are school-level average test scores, region, average class size, and the student-to-teacher ratio (all measured at baseline). A LASSO (Least Absolute Shrinkage and Selection Operator) procedure was used to determine which $X_{igsj} \times POST_t$ interactions to include in the specification.[5] The three-way interaction $\zeta(\eta_j \times \theta_g \times POST_t)$ represents grade-by-matched pair interactions with the post-period indicator. Intuitively, these fixed effects account for how much, on

[5]More specifically, we first calculated each student's change in test scores between the baseline and endline assessments. Then, we residualized this change in test scores, subtracting the comparison group's grade-by-matched pair mean. Lastly, with these residuals, we use a post-double-selection LASSO to select relevant predictors X_{igsj} (Belloni et al., 2011).

average, a PSP student's matched-pair comparison peers (who are enrolled in the same grade in the matched non-PSP schools) learned between baseline and endline; these effects also apply to corresponding PSP students, which in effect implements the parallel trends assumption.

The coefficient of interest is β; it measures the intent-to-treat (ITT) effect of assignment to the program. In estimating ITT effects, we allow for the (plausible) possibility that the program was not fully implemented in at least some of the PSP schools (none of the non-PSP schools received the program). Following de Chaisemartin and Ramirez-Cuellar (2024) on clustering in paired and small-strata experiments, we cluster standard errors at the matched-school-pair level. To account for multiple hypothesis testing, we present (Westfall and Young, 1993) stepdown adjusted p-values, taking into account a prespecified hierarchy of research hypotheses.

For β in Equation (1) to estimate the causal impact of the PSP program on the outcomes of interest, the identifying assumption is that, conditional on the "comparison" variables in that equation, the average trend in the outcomes over time in each PSP school would have been the same as the average trend in that school's matched non-PSP school if the PSP had not been implemented in that PSP school. While this assumption cannot be directly tested, one piece of supportive evidence is shown in Figure 1. It shows that average scores on the end-of-primary exams for the PSP schools and for their matched comparison (non-PSP) schools are very similar for both the Cohort 1 schools and the Cohort 2 schools in the seven school years before the PSP was first implemented in the 2022-23 school year.

4 Data

This study uses two sources of data. The first is administrative data for all of Morocco's public primary schools. The second is baseline and endline assessment data that measure students' performance in mathematics, Arabic and French for 338 public primary schools.

Morocco's Ministry of National Education, Preschools and Sports granted the authors access to historical administrative data on all 8,250 public primary schools[6] in Morocco over a 7-year period, from 2015 to 2022. The administrative data include more than 248 variables measuring student performance and socioeconomic status for 24 million students, as well as many variables on the characteristics of approximately the 104,000 teachers in these schools, and on the characteristics of the 21,077 schools themselves.

This study's main outcomes of interest are student performance on academic assessments in Arabic, French, and mathematics. For the first year of data collection for Cohort 1, baseline assessments for each of these three subjects were administered in September of 2023, and endline assessments were administered in June and July of 2024. For the second year of data collection, baseline assessments for Cohort 2 were administered in September of 2024, and 1-year endline assessments for Cohort 2, along with 2-year endline assessments for Cohort 1, were administered in June and July of 2025. These assessments record students' responses

[6]While there are 8,250 central public primary schools in Morocco, we consider all satellite schools attached to the mother (central) schools for a universe of 21,077 schools.

for all test questions (test "items"). For each subject, a two-parameter logistic (2PL) item response theory (IRT) model (De Ayala, 2013; Kolen and Brennan, 2014) was estimated to aggregate these responses and generate continuous estimates of student performance, using overlapping items (or "anchors') to map test scores onto common scales (across grades as well as across the baseline and endline data collection rounds). In doing so, differential item functioning was investigated, and only test questions with stable item parameters were retained. Each subject's continuous test score was standardized by subtracting the mean, and then dividing by the standard deviation, of the distribution of the test scores at endline of the students in the matched (non-PSP) schools. Further details on the composition and quality of the assessments are elaborated upon in Appendix Section B.

The study's sample of schools was constructed in four steps, using administrative data on the universe of government schools. We performed these steps separately for the Cohort 1 and Cohort 2 samples. The first cohort (Cohort 1) are the original PSP schools and comparison schools selected for the year 1 assessment over the 2023 – 24 academic year. The second cohort (Cohort 2) are the new set of PSP schools and comparison schools selected for the year 2 assessment over the 2024 – 25 academic year. The main difference in these two cohorts of schools are the timing of the rollout of the PSP reforms. The PSP schools in Cohort 1 implemented the program in September of 2023, and so they had two years of exposure to the PSP in July of 2025, while the PSP schools in Cohort 2 implemented the program in September of 2024, and so they had only one year of exposure in July of 2025.

To select these two cohorts of schools, we first used the "post-double-selection" (PDS) methodology (Belloni et al., 2012, 2011, 2014, 2016). For Cohort 1, the four steps were as follows. First, 16 variables were identified that were predictive of either baseline test scores or participation in the program. Second, using these 16 selected variables, Mahalanobis nearest-neighbor matching (without replacement) was used to identify a matched (non-PSP) school for each of the 626 PSP schools that implemented the program in September of 2023. Third, stratifying by region[7], 150 of these PSP schools (along with their 150 non-PSP matches) were randomly selected from the initial 626 PSP schools (and their matched non-PSP schools) for Cohort 1. Fourth, of the 300 Cohort 1 sampled schools (150 PSP and their matched 150 non-PSP schools), 12 schools were removed from the sample due to logistical issues with data collection, their matched counterpart schools were consequently also removed resulting in the removal of 24 schools. Further, for the second year of data collection some Cohort 1 non-PSP schools had to be removed from the sample because they implemented the PSP in September of 2024 and so they were no longer valid comparison schools; their matched PSP schools also had to be removed from the sample since they no longer had a matched non-PSP school. Of the original 276 Cohort 1 schools evaluated over the 2023-24 academic year, only 184 were evaluated in the 2024-25 academic year.

For Cohort 2, the starting point is 35 new PSP schools that had started the PSP in September of 2024; they were selected as the best match pairs of the 2,000 primary schools that started the PSP in that month with the expansion of the PSP and the remaining non PSP schools. We test the balance across school attributes in Table 3. These 35 PSP schools were matched to 35 non-PSP schools using the first three steps of the matching method that was used for the Cohort 1 schools. The fourth step for the 70 Cohort 2 sampled schools (35 PSP and their

[7] In some cases, small regions were grouped with other regions in order to obtain better matches.

matched 35 non-PSP schools), 8 schools were removed from the sample due to logistical issues with four schools, which required the removal of those schools' matched pair schools from the sample. Appendix A provides additional technical details on the sampling of both Cohort 1 and Cohort 2 schools and compares the study's sample of schools with the remaining primary schools in the country, with other program schools, and with the set of matched comparison schools.

The study's unit of analysis is a student. In each school, grade, and subject, surveyors were given two lists of five randomly selected students: a "priority" list and a "secondary" replacement list of students to examine. These lists were constructed before the baseline data collection began, using enrollment records, and they allowed for replacements for priority students who were absent on the day of the baseline assessment. In the first year for Cohort 1, the effective student sample was 20,784 students (6,860 tested in Arabic, 6,858 tested in French, and 7,066 tested in mathematics). In the second year, Cohort 1's effective student sample declined to 11,150 students (3,726 in tested in Arabic, 3,695 tested in French, and 3,729 tested in mathematics). For Cohort 2, the effective student sample size for evaluating the first year of these students' exposure to the program in the 2024 – 25 academic year is 4,053 students (1,302 tested in Arabic, 1,356 tested in French, and 1,395 tested in mathematics). Approximately three-quarters of tested students in both cohorts were drawn from the priority lists.[8] Only students from grade levels 2 through 6 were assessed in the second year 2024 – 2025, to focus on students in Cohort 1 schools who had been exposed to the PSP for two years. In our analysis, we also evaluate a pooled sample across Cohort 1 and 2 for their first year of exposure to the PSP reform.

Appendix C reports on attrition of student observations from baseline to endline assessment across these four samples. We find minimal attrition of student observations after one year, and only modest attrition after two years, with no evidence of differential attrition by treatment status. Table 5 presents the differential attrition rates across treatment status for our four samples. The highest rate of attrition is about 15 percent for the second year follow up for Cohort 1, while the attrition rates of the 1 year exposure samples for Cohorts 1 and 2 are around 5 percent. The differences in rates of attrition between PSP students and comparison school students are not statistically significant; the differential rate of attrition across treatment status are very small, between 1.3 and 2.3 percentage points.

5 Results

ITT effects from one year of exposure. The estimated intent-to-treat (ITT) impacts of the PSP program on student learning are given in Table 1. Across all three subjects (Arabic, French, and mathematics), the PSP education reform increased student performance, relative to students in the comparison (non-PSP) schools, for both Cohort 1 and Cohort 2 in their first year of exposure to the program, as well as for Cohort 1 after two years of exposure to the program. For Cohort 1, after one year of exposure to the PSP students' overall

[8] In Cohort 1 (Year 1), 75% of tested students were on the priority list overall. By subject, the shares were 75% in Arabic, 76% in French, and 75% in mathematics. The proportions were similar in Cohort 2: 77% overall, 76% in Arabic, and 77% in both French and mathematics.

performance (averaged over all three subjects) increased by 0.90 standard deviations (s.d.) of the distribution of endline scores of students in the non-PSP schools (column (1) of Table 1). Cohort 2 had a substantially lower, but still very large overall treatment effect of 0.46 s.d. (column (9)). Figure 2 compares these two ITT effects to those found in the education literature, as reported by Evans and Yuan (2022). For both cohorts, the first year of exposure to the PSP program produced treatment effects that are larger than the 90th percentile of all reported education treatment effects reported in that paper. Finally, column (13) shows pooled estimates (combining both cohorts) of the impact of one year of exposure to the PSP program; the overall estimated ITT effect is 0.68 s.d.

These large effect sizes after one year of exposure to the PSP are seen for all three subjects for both cohorts. The estimated ITT impacts for Arabic are 0.52 s.d. for Cohort 1 and 0.28 s.d for Cohort 2. For French, the estimated ITT impacts are 1.29 s.d. for Cohort 1 and 0.65 s.d. for Cohort 2. For mathematics, the estimated ITT impacts are 0.93 s.d. for Cohort 1 and 0.49 s.d. Cohort 2. For both cohorts, the largest estimated impacts from the PSP reforms were for student performance in French, and the smallest estimated impacts were for Arabic.

In Panel B of Table 1, we estimate the impact of the PSP for male and female students separately. For Cohort 1, the first year of exposure led to a 0.94 s.d. increase in the performance of female students and a 0.86 s.d. increase for male students, averaging over all subjects. For Cohort 2, the first year of exposure to the PSP program resulted in a 0.51 s.d. increase in female students' performance and a 0.41 s.d. increase in male students' performance, again averaging over all subjects. This suggests that female students benefit more from the PSP intervention than male students, but these differences are not very large relative to the overall impact of the PSP on both male and female students.

Panel C of Table 1 compares the effect of the PSP by baseline performance. We consider the effect for students in the bottom quartile (lowest 25 percent) of baseline student performance and for students in the top quartile (highest 25 percent) of baseline student performance. This is done to investigate whether the impact of the PSP intervention varies by students' initial levels of learning. Reassuringly, we find that the PSP ITT effects are very strong for students in both quartiles. For Cohort 1, the students in the bottom quartile (of baseline student performance) had an overall ITT impact of 0.95 s.d., while the students in the top quartile had an overall ITT impact of 0.84 s.d. In Cohort 2, the students in the bottom quartile had an overall ITT impact of 0.47 s.d., while those in the top quartile had an overall ITT impact of 0.38 s.d. These results suggest that lower performing students may benefit the most from the PSP education reform. Yet top performing students continue to benefit significantly from the PSP education reform as well.

ITT effects from two years of exposure. Columns (5) - (8) of Table 1 present estimates of the impact of the PSP for Cohort 1 after two years of exposure to this program. The estimated impacts after two years of exposure to the PSP are somewhat smaller than the estimated impacts from the first year of exposure. Across all subjects, the ITT effect after two years is 0.59 s.d., averaging over all three subjects. The ITT estimates by subject have the same pattern as the estimates after one year. Students' estimated impacts for Arabic had the lowest ITT effect (0.34 s.d.), students' estimated impacts for French were the highest (0.91 s.d.), and the estimated impact for mathematics was in between these two estimates

(0.59 s.d.). For female students, the overall estimated ITT effect is 0.63 s.d., compared to 0.55 s.d. for male students, again suggesting that the PSP program benefited female students more than male students. By baseline performance, the bottom quartile of students had an estimated ITT effect of 0.55 s.d., compared to a 0.60 s.d. for the students in the top quartile, although this difference is not statistically significant. While this differs from the findings for the first year of exposure to the PSP program for Cohort 1, it is still the case that the PSP program has large impacts on both the top and the bottom quartiles.

Grade level effects. Sub-analysis of these ITT effects by grade level are presented in Table 6 of Appendix D. For both cohorts, all grades benefited from the PSP program, although these estimates suggest that the impacts were somewhat higher for grades 4 – 6 than they were for grades 2 and 3.

Heterogeneous effects by baseline performance. To address concerns that the PSP may be less effective or even disadvantageous for initially high-performing students, we estimate heterogeneous ITT effects along the baseline achievement distribution. For each subject and the overall score, we convert baseline scores into percentiles and re-estimate the main ITT specification locally at each percentile using triangular kernel weights centered at that point, with a bandwidth of one-quarter of the percentile distribution. This yields a smooth estimate of treatment effects by baseline performance.[9] Figures 3 and 4 plot these ITT curves with point-wise 95% confidence intervals; estimates are less precise at the extreme tails since the effective sample sizes are smaller.

Across cohorts and exposure lengths, impacts remain strongly positive throughout the distribution. We focus here on the upper tail. For Cohort 1 after two years of exposure (Figure 3), estimated effects on the overall score and each subject remain positive among top-performing students. The overall effect tends to rise with baseline achievement, driven primarily by stronger gains in French and mathematics, while effects on Arabic scores are flatter but still positive over most of the upper tail. Although precision declines at the very top percentiles, as expected given fewer effective observations, these results indicate that higher-performing students benefit almost as much as their peers, and in one case (French) even more.

For Cohort 2 after one year of exposure (Figure 4), we find similar ITT patterns along baseline score distribution. The impacts for high-baseline students are positive across subjects and on the overall score, though confidence intervals are wider due to the smaller number of schools. For the very top performing students, the impacts of the PSP on French skills appear somewhat higher than the impacts of all other students, except for the very highest performers. For Arabic and mathematics, most of the top students scored somewhat lower, although at the very top the estimated impacts are somewhat higher (although these are very imprecisely estimated). Combining all scores, top performers seem to do slightly worse than average, but they have certainly gained from the program. Taken together, these results provide no evidence that the PSP has a negative effect on initially high-performing

[9] This is akin to a quantile regression but relative to a specific covariate dimension.

students; if anything, impacts at the top of the distribution are comparable to – and for French, often larger than – those for initially lower-baseline students.[10]

Sub-domains of student learning. Sub-analysis of the ITT effects by the sub-domains of student learning are shown in Table 7 of Appendix D. Across all subjects, we explore the effect for exam items which are below the current grade level of the student, as well as exam items which are at the current grade level. Overall, we find no large differences between these two types of test items; the PSP increases student performance on both types of items to a similar extent. For Arabic and French, we also examine how many words in each language the student correctly identified. For French, exposure to the PSP increases the number of words correctly answered for all three sets of students (Cohort 1 after 2 years, Cohort 2 after 1 year, and both cohorts pooled after one year). For Arabic, the same holds for Cohort 1 after two years and for the pooled sample after one year, but there is no significant effect for Cohort 2 after one year. Finally, for mathematics, we explore student performance on four subsets of mathematical learning: calculation and numeracy; geometry, measures, and data; identification knowledge; and application and reasoning. The effects are somewhat larger for calculation and numeracy and for identification knowledge. Further analysis of these differences may be valuable for understanding the reasons for the PSP's large impact on students' mathematical skills.

6 Mechanisms

An important finding from this paper is that, for Cohort 1 students, who have experienced two years of exposure to the PSP, the ITT effects are smaller in the second year compared to the first year. In this section, we propose a ceiling effects as a minor mechanism driving this reduction in the ITT over time for some French test performance. In Appendix Section E, we present an event study analysis exploring the impact of time dynamics influencing the PSP impacts. In Appendix Section F, we also present analysis using only anchor item test questions to rule out the influence of changing test questions over time as a potential explanation. In Appendix Section G, we present several other untestable hypotheses which may contribute to this result.

The reduction in the impact of the program in the second year for Cohort 1 students means that the students in comparison schools are catching up with PSP students. There is one hypothesis as to why this occurs which we can test using our data: the IRT measures of student performance may have an upper bound, creating ceiling effects. In essence, high performing students in the PSP schools may already been answering all the questions correctly so they cannot increase their scores further (e.g., they have reached the ceiling of the skills measured by the IRT), while there is still room for almost all of the students in the non-PSP schools to increase their scores. Ceiling effects are applicable to any measurement of student skills, both summary test scores and performance as measured by IRT scores.

[10]Similar patterns are observed for Cohort 1 after one year of exposure (Figure 9) and in the pooled sample (Figure 10).

To determine whether there are ceiling effects limiting the progress of IRT test performance, we examine the change in the distribution of IRT scores by treatment group over time. Figure 5 presents the probability density functions (PDFs) of IRT scores by survey wave and by subject. For this figure (and subsequent analysis), we restrict the sample only to students who have completed all three exams (baseline in year 1, endline in year 1, and endline in year 2). Therefore the changes in test performance are not due to changes in the composition of students across exams or time. The IRT performance of students at baseline is shown in red, while blue shows students' IRT performance on the year 1 endline exams, and green shows their IRT performance on the year 2 endline exams. Comparison school students are represented by the dashed lines, while the PSP students are represented by the solid lines.

Immediately apparent is that comparison school students and PSP students performed very similarly at baseline. Then, at the end of the first year, both comparison and PSP students increased their test performance, but PSP students increased their performance much more than did the comparison school students. This figure visually demonstrates the results in columns (1) – (4) of Table 1. This is evident by the rightward shift of the blue PDF curves, with the solid (PSP) curve clearly farther to the right of the dashed (comparison) curve. This shows that the PSP students had much higher IRT performance than the comparison school students. At the end of the second year (green lines), the comparison students (dashed) performed somewhat better than they did at the end of the first year (the distributions in the dashed green lines are slightly to the right of the distributions in the dashed blue lines). But the PSP students' (solid) PDF curves have not moved to the right (the solid green lines are not to the right of the solid blue lines); in fact, they have shifted slightly to the left. This suggests that PSP students are not performing better in the second year than in the first year. Indeed, they may have performed slightly worse in the second year, although they still perform better than the comparison school students in all the three subjects (Arabic, French, and mathematics) at the end of year 2. Overall, this indicates that the smaller ITT effects in year 2 are due to comparison school students increasing their performance in year 2, while the PSP students did not increase their performance in that year, and may have performed slightly worse. This suggests that the PSP students are benefiting much less, and perhaps not at all, from that program in the second year than in the first year, while the comparison school students improved their performance in both year 1 and year 2.

To complement our investigation of whether there are ceiling effects that prevent PSP students from continuing to outperform comparison school students on IRT scores, we constructed Figure 6, which presents the cumulative distribution function (CDF) curve for IRT performance over all three testing periods (baseline, end of year 1 and end of year 2) for overall composite test scores. The colors and line types are the same as in Figure 5. In this figure, the farther to the right a line is, the smaller is the percentage of students who score at that IRT score. As an example of how to read this figure, at an IRT score of 0 the comparison students in year 2 (dashed green line) have roughly 40 percent of students performing with an IRT score of 0 or below. To detect the presence of ceiling effect, we are visually looking for a vertical jump in the CDF curve on the right tail of the lines. This represents a large portion of students all scoring the highest possible IRT score. For the PSP students (solid blue and green lines), we observe a modest jump at the tail end of IRT scores near 3. One way to detect ceiling effects is to look at the density plots in Figure 5. If the right tails of

those distributions do not come all the way down to the horizontal axis, this indicates that a noticeable proportion of the students are at the upper bound of the test score. This is clearly evidence for PSP students' French IRT scores in years 1 and 2, and to a lesser extent for their mathematics IRT scores in year 1. This also shows up for both years in the overall score.

However, these ceiling effects do not affect most PSP students, so they cannot explain the lack of improvement in student learning for the PSP students from year 1 to year 2. There is some evidence that French test performance as some minor ceiling effects, but this is not present for either Arabic or mathematics test performance. So the reduction in the PSP impact from year 1 to year 2 remains unexplained. Further research is needed to understand why the impact of the PSP is lower in year 2 than in year 1.

7 Conclusion

The report by Ibrahim et al. (2025) found impressively large impacts of Morocco's PSP education reform on student learning after one year. This report examines the impacts of the PSP program after two years. More specifically, it estimates impacts of two years of the PSP program for the schools that started the program in September of 2023 (which we call the Cohort 1 schools), and it also estimates impacts after one year for a new set of primary schools that started that program in September of 2024 (which we call Cohort 2 schools).

We find that the impacts of one year of the PSP for the Cohort 2 schools are very high (see Figure 2), but the impacts are only about half the size of the impacts after one year in the Cohort 1 schools. The reasons for this require further investigation, but it could reflect the fact that school participation in the PSP is voluntary; it is likely that the Cohort 1 schools, which were the first to volunteer to implement the PSP, are the schools that were the most enthusiastic about implementing that program, perhaps because they understood that the program was likely to have large impacts in their schools. With the expansion of the reform to a larger set of schools in Year 2, the PSP is likely to have had a smaller impact in the second year.

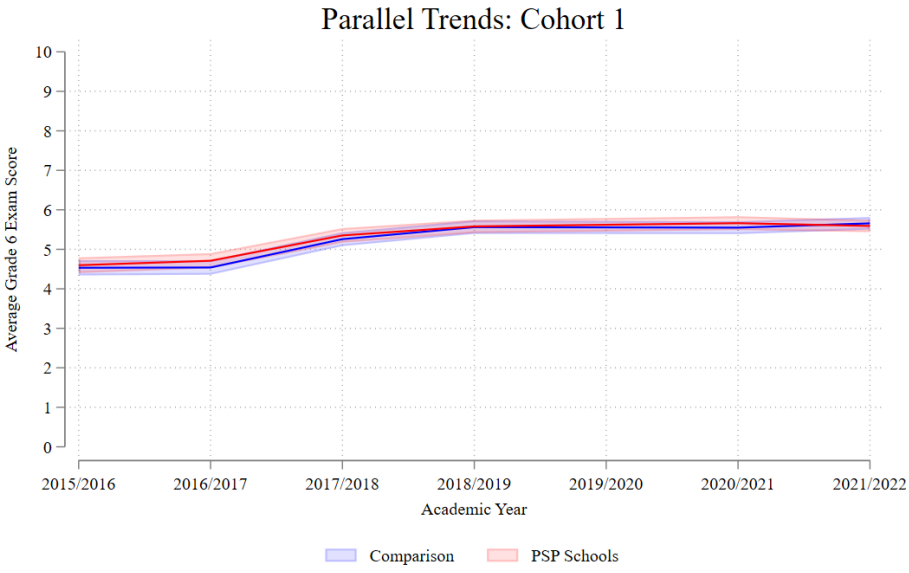
More interesting is the finding that the impact of the PSP program on the Cohort 1 schools is lower after two years than it was after one year. We are continuing our exploration of the shift in ITT effects from year 1 to year 2. While one would expect that the additional benefits from the second year of the program would not be as large as the initial gains experienced in the first year, a decline in the impact over time is somewhat surprising. The investigations in the mechanisms section (Section 6) have thus far not found a clear explanation, and more research is needed to explain the potential avenues through which the shift in ITT operates. Administrative data on the implementation of the PSP in both years could be very useful for understanding the reduced impact of the program in year 2.

The following possible explanations require further investigation, and require more data. First, it may be that the matched non-PSP schools are "catching up" to the PSP schools for reasons unrelated to the program, such as better implementation of existing policies in those schools. A second, related possibility, which needs to be checked using administrative data, is that some of the non-PSP schools have found a way to adopt some the the PSP interventions. Specifically, evidence of teacher transfers from PSP schools to comparison

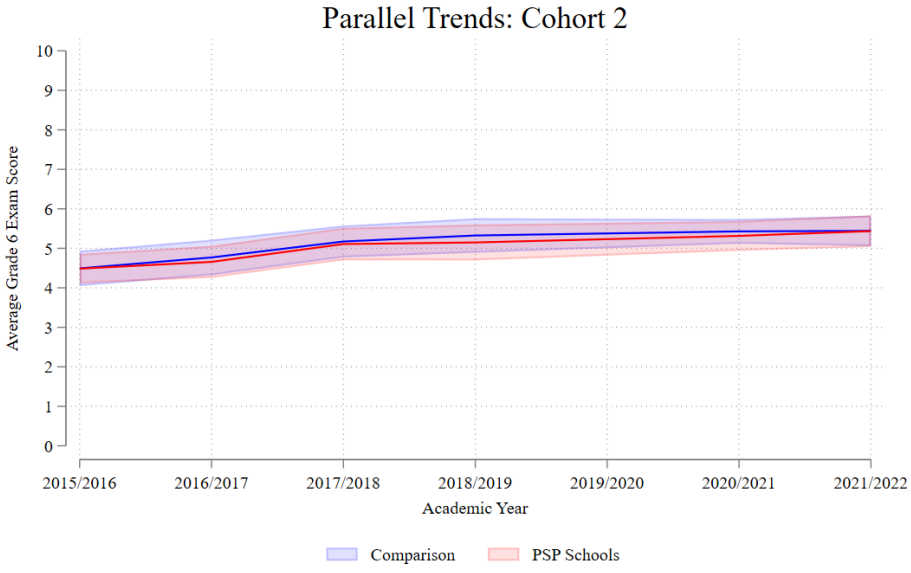
schools or classroom observation of the pedagogy conducted in comparison schools (e.g., whether comparison school teachers are implementing any PSP practices) would be helpful to investigate this issue. Additionally, the scale up of the PSP from 626 schools in the first year to 2,626 schools in the second year may impact the effectiveness of the PSP. With a wider set of schools to train and implement the reform in, the scale up of the PSP may have diluted the more concentrated impact of program for the first year. Finally, there may be some "slowing down" or even a partial reversal of implementation of the PSP in the 626 Cohort 1 schools. Perhaps the initial enthusiasm declined over time, or the very strong impacts in the first year resulted in a sense of complacency in the second year that reduced teacher effort. We are aware that for example PSP teachers in Year 1 were aware that they would receive a financial compensation for implementing PSP, which following nation-wide teacher strikes that happened in (add month and year), was restructured to a blanket salary increase for all teachers nation-wide. In Year 2, PSP teachers hence did not have a specific financial incentive to implement PSP. Further research on the mechanisms driving these effects, and whether they will intensify with the continued expansion of the PSP program nationally, which would benefit from the collection of new types of data (e.g., classroom implementation, teacher surveys, etc.), is needed to understand the reasons why the extremely high learning gains in the first year in the Cohort 1 PSP schools seem to have fallen in the second year of the program for those schools.

Exhibits

Figure 1: *Parallel Trends of Test Scores Across Treatment Assignment by School*



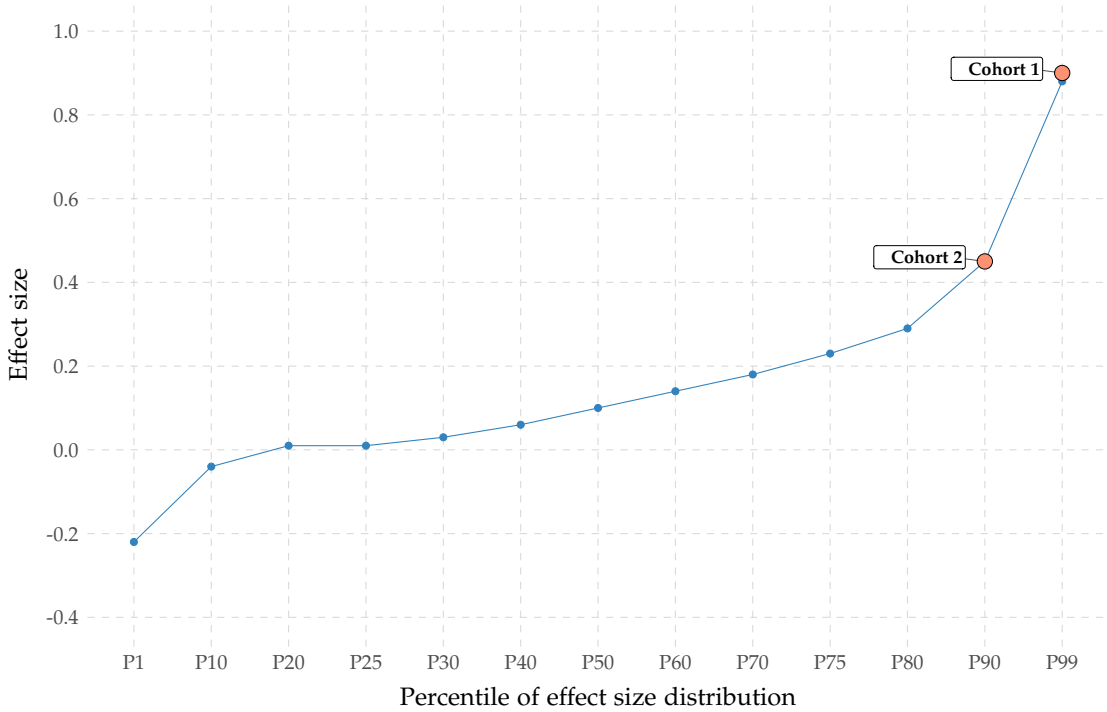
(a) *Parallel Trends for Cohort 1 Sample*



(b) *Parallel Trends for Cohort 2 Sample*

Note: This figure confirms that the program and comparison schools exhibited similar levels and similar trends in exam scores before the program started (2022/23). It presents, for the 1,699 PSP program schools and 1,699 comparison schools, the yearly average end-of-primary exam scores over a period of seven years before the program launched.

Figure 2: *Comparison of ITT Effects to Impacts Observed in Other Countries*



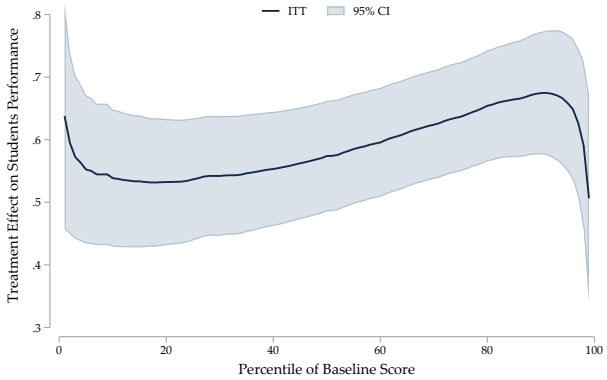
Note: This figure shows the distribution of effects reported in Evans and Yuan (2022). Cohort 1 reflects the program's impact on the overall composite IRT score after one year of exposure in a sample of 276 schools, while Cohort 2 captures the effect size in the second year based on a sample of 62 schools.

Table 1: *ITT Effect of PSP Program on Students Performance (IRT Scores)*

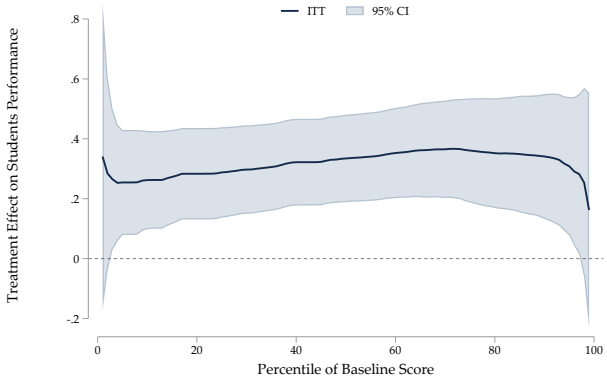
	1-Year Exposure (Cohort 1)				2-Year Exposure (Cohort 1)				1-Year Exposure (Cohort 2)				Pooled Exposure (Cohort 1 & 2)			
	Overall (1)	Arabic (2)	French (3)	Math (4)	Overall (5)	Arabic (6)	French (7)	Math (8)	Overall (9)	Arabic (10)	French (11)	Math (12)	Overall (13)	Arabic (14)	French (15)	Math (16)
Panel A: Overall																
All students	0.899***	0.523***	1.291***	0.926***	0.590***	0.338***	0.909***	0.591***	0.457***	0.279**	0.647***	0.490***	0.681***	0.404***	0.962***	0.703***
	(0.030)	(0.048)	(0.048)	(0.036)	(0.041)	(0.069)	(0.051)	(0.054)	(0.062)	(0.116)	(0.069)	(0.069)	(0.040)	(0.062)	(0.055)	(0.045)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.023]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Panel B: By gender																
Female	0.940***	0.514***	1.335***	0.931***	0.634***	0.341***	0.963***	0.575***	0.507***	0.416***	0.623***	0.542***	0.727***	0.469***	0.974***	0.730***
	(0.033)	(0.053)	(0.052)	(0.045)	(0.045)	(0.075)	(0.064)	(0.067)	(0.067)	(0.129)	(0.070)	(0.085)	(0.043)	(0.068)	(0.059)	(0.053)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.003]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Male	0.860***	0.523***	1.219***	0.900***	0.554***	0.296***	0.841***	0.609***	0.412***	0.164	0.647***	0.440***	0.637***	0.340***	0.923***	0.669***
	(0.034)	(0.057)	(0.058)	(0.046)	(0.045)	(0.087)	(0.063)	(0.067)	(0.065)	(0.127)	(0.084)	(0.069)	(0.043)	(0.072)	(0.062)	(0.048)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.001]	[0.000]	[0.000]	[0.000]	[0.207]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Panel C: By baseline performance																
Bottom quartile	0.948***	0.627***	1.229***	0.849***	0.549***	0.345***	0.735***	0.588***	0.468***	0.308*	0.597***	0.461**	0.707***	0.486***	0.905***	0.646***
	(0.055)	(0.079)	(0.111)	(0.082)	(0.060)	(0.126)	(0.149)	(0.082)	(0.107)	(0.152)	(0.137)	(0.172)	(0.062)	(0.082)	(0.095)	(0.098)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.007]	[0.000]	[0.000]	[0.000]	[0.052]	[0.000]	[0.012]	[0.000]	[0.000]	[0.000]	[0.000]
Top quartile	0.835***	0.414***	1.152***	0.977***	0.604***	0.258**	0.881***	0.541***	0.383***	0.268	0.708***	0.283*	0.610***	0.327***	0.933***	0.625***
	(0.035)	(0.070)	(0.086)	(0.062)	(0.056)	(0.125)	(0.091)	(0.096)	(0.079)	(0.181)	(0.119)	(0.140)	(0.048)	(0.104)	(0.075)	(0.089)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.042]	[0.000]	[0.000]	[0.000]	[0.150]	[0.000]	[0.053]	[0.000]	[0.002]	[0.000]	[0.000]
Number of Students	20,784	6,860	6,858	7,066	11,150	3,726	3,695	3,729	4,053	1,302	1,356	1,395	24,837	8,162	8,214	8,461
Number of Schools	276				184				62				338			

Note: This table reports on the program's intent-to-treat (ITT) effects among all students (Panel A), by gender (Panel B), and students' baseline learning level as per the distribution within their enrolled grade level (Panel C). The outcome of student performance is measured as the latent ability parameter α from a two-parameter Item Response Theory (IRT) model. Each subject level exams (Arabic, French, and Math) is reported alongside a composite score stacking the three sub-samples. Standard errors are clustered at the matched-pair level and shown in parentheses. Adjusted p-values using the Westfall–Young step-down procedure are reported in brackets. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

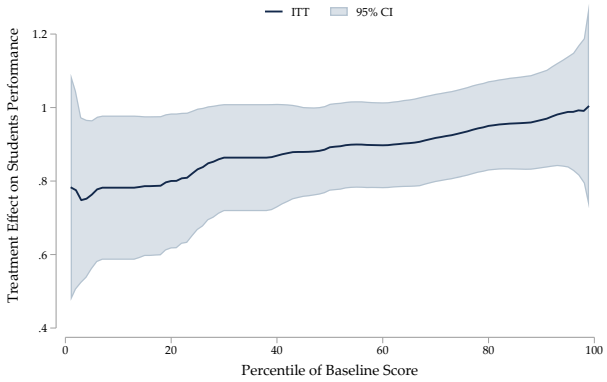
Figure 3: *Cohort 1: ITT Effects by Baseline Achievement Percentile After Two Years of PSP Exposure*



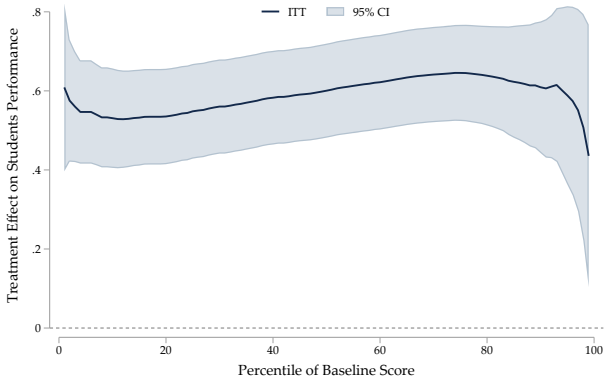
Overall



Arabic



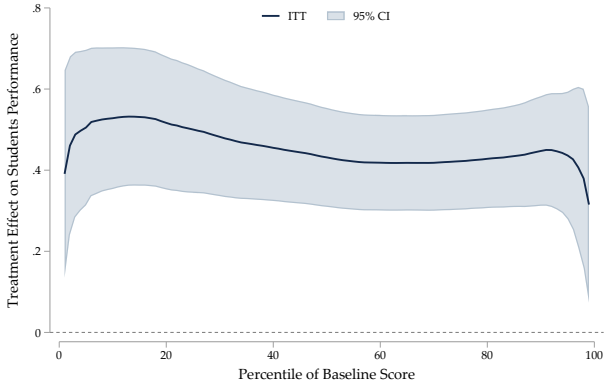
French



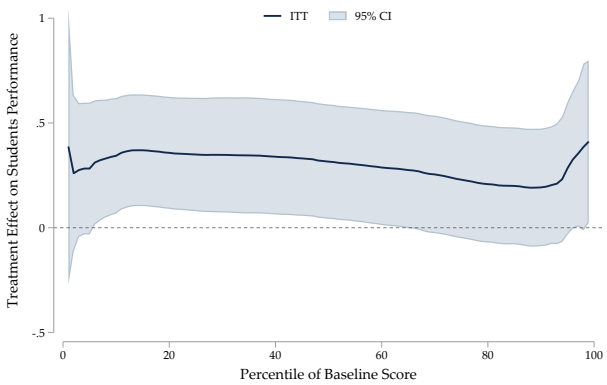
Math

Note: This figure shows the ITT effect at each Quantile of the baseline test score distribution. This is a form of local rolling regression. The black line is the ITT effect and the shaded gray area are the 95 percent confidence interval for the estimated effect. This figure only estimates the effects for the Cohort 1 sample after two years of exposure to the PSP.

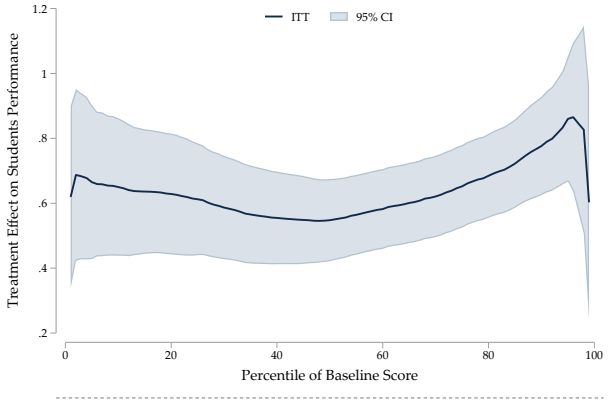
Figure 4: *Cohort 2: ITT Effects by Baseline Achievement Percentile After One Year of PSP Exposure*



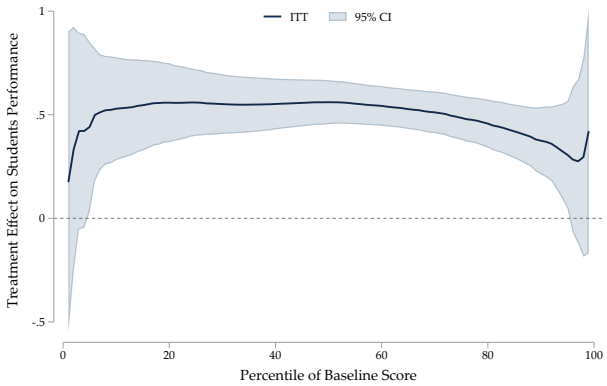
Overall



Arabic



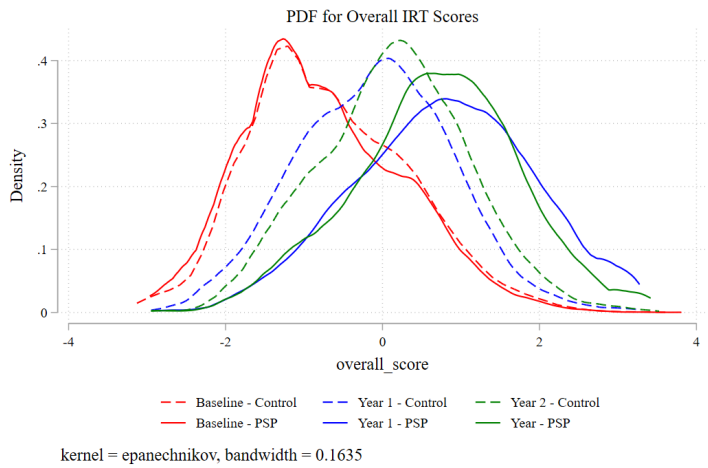
French



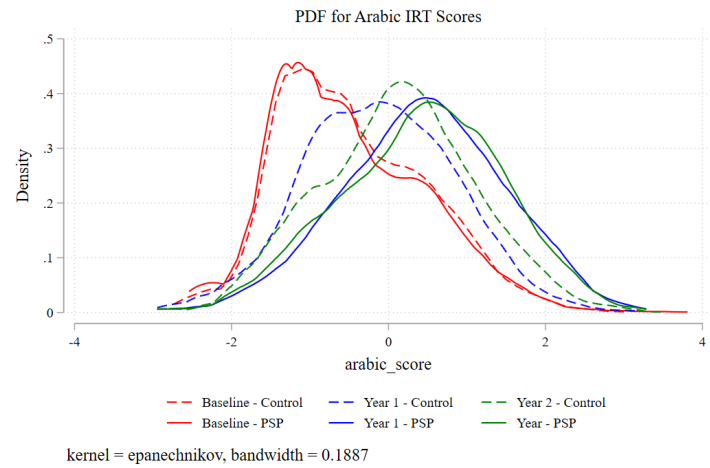
Math

Note: This figure shows the ITT effect at each Quantile of the baseline test score distribution. This is a form of local rolling regression. The black line is the ITT effect and the shaded gray area are the 95 percent confidence interval for the estimated effect. This figure only estimates the effects for the Cohort 2 sample after one year of exposure to the PSP.

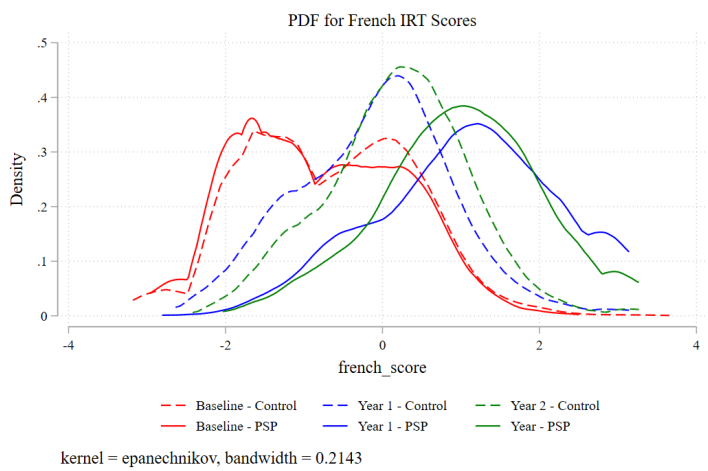
Figure 5: *PDF Density of IRT Test Performance by Survey Wave*



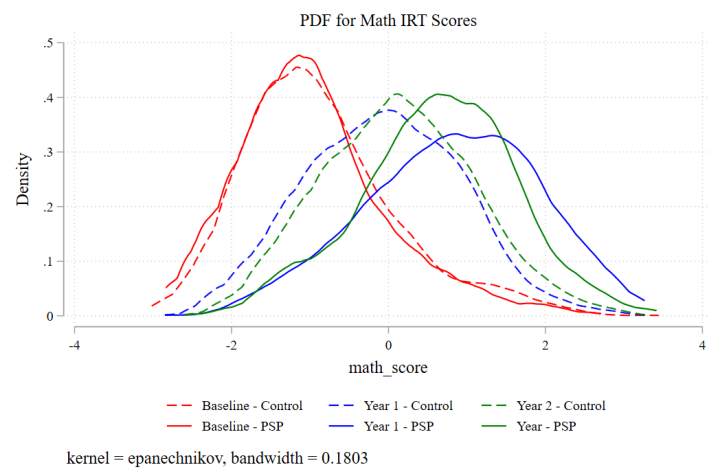
(a) *Overall*



(b) *Arabic*



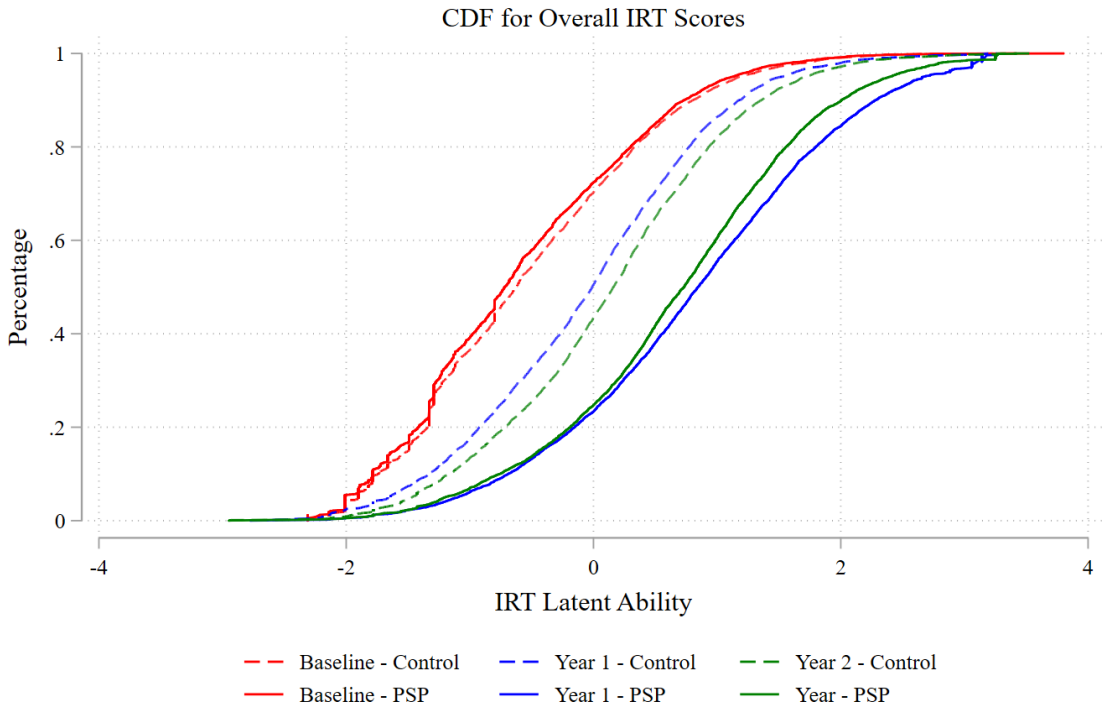
(c) *French*



(d) *Math*

Note: These four figures present the probability density function (PDF) for the composite IRT scores by the wave of the exam. Red represents the baseline exam. Blue represents the endline exam after 1 year. And green represents the endline exam after 2 years. The dashed lines represent students in comparison schools, and solid lines represent students in PSP schools. Sub-figure A shows the overall IRT scores. While sub-figures B, C, and D show the Arabic, French, and math subject IRT scores, respectively. To estimate the density function, we use a Epanechnikov (parabolic) kernel.

Figure 6: *CDF of Overall IRT Test Scores by Year*



Note: This figure shows the cumulative distribution function (CDF) for the IRT latent test performance as a composite overall score by the wave of the exam. Red represents the baseline exam. Blue represents the endline exam after 1 year. And green represents the endline exam after 2 years. The dashed lines represent students in comparison schools, and solid lines represent students in PSP schools.

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Supplementary Appendix

A School Representativeness

To improve the viability of the conditional parallel trends assumption, we matched schools using the best predictors of treatment status and baseline outcome (Ham and Miratrix, 2024). With access to a large administrative dataset on all primary schools in Morocco, we were able to implement a machine learning algorithm to identify the most relevant predictors of treatment status and a baseline measure of the outcome of interest (Ham and Miratrix, 2024). The algorithm used is post-double selection LASSO (PDSLASSO) used for estimating structural parameters in linear models with many controls (Belloni et al., 2012, 2011, 2014, 2016) .

The PDSLASSO algorithm operates by performing two LASSO regressions. First, it estimates a LASSO regression with the baseline outcome variable as the dependent variable and a set of potential comparison variables as regressors (all 248 constructed variables in the database). Second, it performs another LASSO regression with the treatment indicator as the dependent variable and the same set of comparison variables as regressors. The final set of comparison variables included in the model is the union of the variables selected in these two steps. This approach ensures that the chosen controls are those most predictive of both the outcome and the treatment assignment, thereby enhancing the robustness of causal inference (Ahrens et al., 2019).

Using a school-level database with 248 variables, consisting of school-level characteristics and aggregated student and teacher characteristics in 8,034 primary schools (including the pre-selected program schools), the PDSLASSO implemented identified 16 variables to be used for matching.

Next, we implemented an iterative process of a calibrated Mahalanobis distance matching algorithm using the variables selected by the PDSLASSO, to match untreated schools to the selected PSP schools for treatment. After each iteration of matching with different distance calipers, we implemented post-matching diagnostics, including tests for common support, external validity, and balance between the treated and the matched untreated schools. External validity was investigated by comparing the probability density function (PDF) and the cumulative distribution (CDF) for the average baseline measure of the outcome of interest in matched, unmatched, and all schools. The CDF is presented in Figure 7 as well as a corresponding boxplot in Figure 8. Balance in baseline characteristics between the matched treated and untreated schools was evaluated by checking for any statistically significant difference between the paired schools.

The Mahalanobis distance matching was implemented without replacement while restricting the algorithm to select one matched (nearest neighbor) untreated school from the full list of untreated schools. Initially, without distance calibration, the matching algorithm successfully matched treated schools with untreated schools. However, the matched untreated schools from this uncalibrated attempt had significantly different characteristics compared to their matched treated schools, based on the same variables selected by the PDSLASSO. To address this imbalance, we iteratively adjusted the Mahalanobis distance caliper in descending

order, starting with the highest possible caliper. We evaluated the baseline balance between schools and examined the PDF and CDF of the baseline measure of the outcome of interest. This process continued until we identified a caliper that produced matched schools with balanced baseline characteristics.

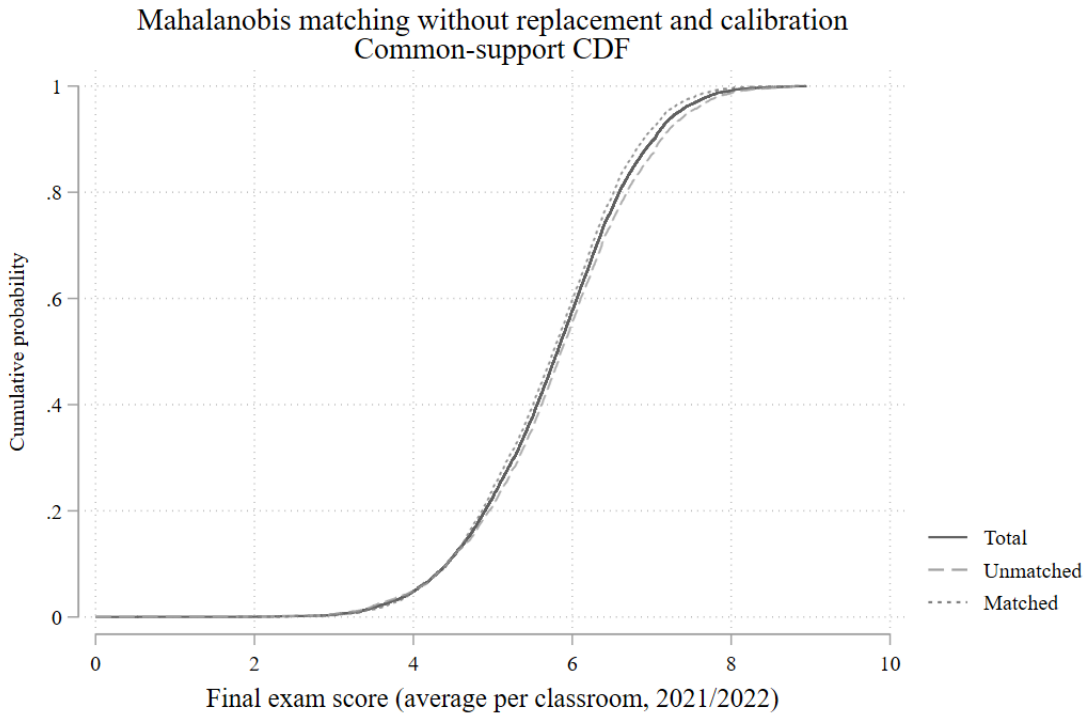
The final caliper used in the matching exercise was root 11.5, producing a list of matched pairs of schools. The population of good matches from the previous step was then used to randomly sample the schools to participate in the baseline and endline data collection for the evaluation of the Pioneer School Program. The evaluation sample was randomly drawn, stratified by terciles of the final grade 6 exam score (the measure for the baseline outcome of interest) and the 12 regions in Morocco, resulting in a total of 36 strata. However, there were 6 strata belonging to the 3-baseline outcome terciles in 2 regions with very few schools. As a result, we only stratified the school sampling by region for those 2 regions, with a final total of 30 strata.

A total of 248 schools were planned for testing at endline — 184 from Cohort 1 and 64 from Cohort 2. In practice, 247 schools were tested. One treatment school from Cohort 2 could not be assessed due to logistical constraints, and to maintain the integrity of the experimental design, its paired comparison school was also excluded. Consequently, the final sample included 62 schools in Cohort 2. The initial endline dataset contained 15,581 student records collected across all participating schools. Several cleaning steps were undertaken to ensure that the final analytic sample included only valid, complete, and unique observations. First, 79 students were removed because their assessments were incomplete or partially filled, including those who did not finish the test or whose forms contained missing responses.

Next, 8 duplicate submissions were dropped based on enumerator feedback identifying the valid version. For students with multiple submissions but no feedback, only the most recent submission was retained, leading to the removal of 47 additional records. A final duplicate review identified further repeated entries; keeping only the most recent submission for each student resulted in 124 additional records being dropped. At the school level, the treated school previously mentioned that was not observed at endline, along with its paired comparison, was excluded from the dataset. This resulted in the removal of 35 student records associated with these two schools.

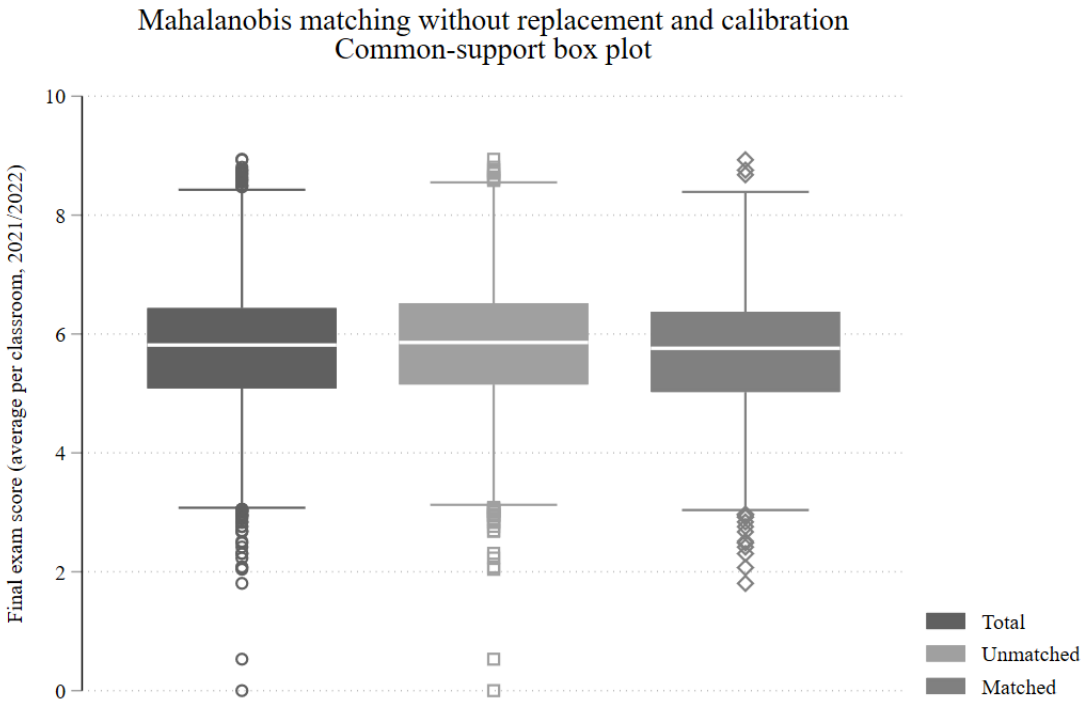
After all exclusions, the final cleaned dataset included 15,288 students across 247 schools, representing the complete and valid sample used for analysis.

Figure 7: *CDF of Common Support Across Treatment Assignment by School*



Note: This figure shows a cumulative distribution function (CDF) for the common support for the most recent final exam score (2021/22) across schools that are selected for the matched pair design (in study) and those unmatched (not in study).

Figure 8: *Common Support Across Treatment Assignment by School for Average Exam Score*



Note: This figure shows a box plot of the common support for the most recent final exam score (2021/22) across schools that are selected for the matched pair design (in study) and those unmatched (not in study).

Table 2: *Cohort 1 Balance of Sample Schools and Representativeness*

	Representativeness (Overall)			Representativeness (within Program)			Balance	
	Non-Study	Study	Difference	Non-PSP Schools	PSP Schools	Difference	Comparison	Difference
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Number of teachers	19.52	7.98	11.53***	19.38	17.62	1.76*	19.51	-0.28
	[9.91]	[8.41]	(0.51)	[9.97]	[10.07]	(0.97)	[9.89]	(1.76)
Urban	0.71	0.15	0.56***	0.68	0.58	0.11**	0.72	0.00
	[0.46]	[0.36]	(0.02)	[0.47]	[0.49]	(0.05)	[0.45]	(0.12)
Total enrollment (2021/2022)	534.68	197.29	337.40***	528.65	485.37	43.28	535.42	-43.85
	[314.97]	[256.61]	(15.60)	[305.65]	[320.50]	(30.65)	[325.69]	(75.26)
Female students (percentage, 2021/2022)	48.17	48.05	0.12	48.16	48.17	0.00	48.12	-0.86
	[3.08]	[7.32]	(0.44)	[3.23]	[5.13]	(0.46)	[2.94]	(1.13)
Qualified for Tayssir (percentage, 2021/2022)	17.70	56.22	-38.52***	17.94	25.24	-7.30**	18.12	-1.19
	[28.36]	[29.17]	(1.77)	[27.34]	[31.20]	(2.94)	[29.74]	(7.09)
Qualified for social security (percentage, 2021/2022)	17.81	56.58	-38.77***	18.02	25.45	-7.43**	18.25	-1.19
	[28.53]	[29.34]	(1.78)	[27.36]	[31.51]	(2.96)	[30.04]	(7.09)
Average final grade-6 score (2021/2022)	5.62	5.74	-0.13*	5.59	5.67	-0.07	5.65	0.01
	[1.00]	[1.14]	(0.07)	[1.01]	[1.05]	(0.10)	[0.99]	(0.21)

Note: This table reports on the study's sample of schools for Cohort 1. Study refers to the 300 sampled schools (150 PSP and their matched 150 non-PSP schools). Non-study refers to all other government primary schools in Morocco. Non-PSP schools refer to the 150 comparison schools not involved in the PSP curriculum reform, and PSP schools refers to the 150 treated schools selected for the PSP curriculum reform. None of the measured differences on observable school characteristics are statistically significant.

Table 3: *Cohort 2 Balance of Sample Schools and Representativeness*

	Representativeness (Overall)			Representativeness (within Program)		
	Non-Study	Study	Difference	Non-PSP Schools	PSP Schools	Difference
	(1)	(2)	(3)	(4)	(5)	(6)
Final exam score (average per classroom, 2021/2022)	5.701	5.694	0.007	5.443	5.436	0.007
	[0.023]	[0.023]		[0.197]	[0.204]	
Final exam score (average per classroom, 2020/2021)	5.516	5.559	-0.043	5.429	5.315	0.114
	[0.024]	[0.025]		[0.159]	[0.192]	
Final exam score (average per classroom, 2018/2019)	5.424	5.417	0.006	5.325	5.149	0.176
	[0.026]	[0.027]		[0.222]	[0.232]	
Final exam score (average per classroom, 2017/2018)	5.230	5.228	0.002	5.173	5.108	0.064
	[0.027]	[0.028]		[0.206]	[0.209]	
Final exam score (average per classroom, 2016/2017)	4.591	4.567	0.024	4.769	4.658	0.112
	[0.029]	[0.031]		[0.229]	[0.206]	
Final exam score (average per classroom, 2015/2016)	4.446	4.470	-0.024	4.492	4.486	0.007
	[0.030]	[0.030]		[0.228]	[0.191]	
Urban	0.597	0.603	-0.006	0.629	0.629	0.000
	[0.012]	[0.012]		[0.083]	[0.083]	
Number of Teachers	18.167	18.297	-0.130	18.800	18.743	0.057
	[0.242]	[0.233]		[1.646]	[1.654]	
School Type: Community	0.044	0.044	0.000	0.029	0.029	0.000
	[0.005]	[0.005]		[0.029]	[0.029]	
School Type: Central/Sector	0.309	0.303	0.006	0.286	0.286	0.000
	[0.011]	[0.011]		[0.077]	[0.077]	
Multi-level classrooms (share, 2021/2022)	0.056	0.051	0.005	0.055	0.078	-0.023
	[0.004]	[0.004]		[0.031]	[0.031]	
Average number of Students Repeating (2015/2016)	3.145	3.128	0.017	3.023	2.900	0.123
	[0.036]	[0.038]		[0.234]	[0.235]	
Tayssir Beneficiary	11.175	11.595	-0.420	11.057	10.509	0.548
	[0.269]	[0.268]		[1.819]	[1.980]	

Note: This table reports on the study's sample of schools for Cohort 2. Study refers to the 70 schools included in the Cohort 2 study's effective sample. Non-study refers to all other government primary schools in Morocco. Non-PSP schools refer to the 35 control schools not involved in the PSP curriculum reform, and PSP schools refers to the 35 treated schools selected for the PSP curriculum reform. None of the measured differences on observable school characteristics are statistically significant.

B Survey Instrument

We administer four exam surveys to measure student performance over two years. Two of these are baseline exams to measure baseline performance separately for Cohort 1 and Cohort 2. The remaining two exams are endline exams to measure the progression in student test performance at the end of the first and second year of the evaluation period. Table 4 presents the number of question items for each exam administered at each grade level across the three subjects. Test items were constructed in coordination with the Ministry of Education. Examines were administered by enumeration staff provided by the Ministry of Education with permission from school administrations. Notably, in the second year we do not test grade 1 performance.[11] Additionally, the majority of French subject exams were shared across grade levels.

To assess instrument quality, we examined item difficulty and discrimination for each subject-by-grade test form using a two-parameter logistic (2PL) item response theory (IRT) model. "Don't know" responses were scored as incorrect in the main specification, consistent across all four survey exams.

Across subjects, item difficulties covered a broad range, indicating that the tests included both easy and challenging items rather than concentrating at a single difficulty level. In mathematics, difficulty parameters ranged from 2.36 to 2.33 and discrimination parameters from 0.34 to 3.63, confirming that the instrument appropriately captured the proficiency range expected across grades. For Arabic, difficulty ranged from 2.51 to 3.60 and discrimination from 0.30 to 4.23, showing broad coverage of student ability levels with most items providing meaningful differentiation and a small subset of weakly discriminating items. The French test exhibited difficulty parameters between 1.99 and 0.90 and discrimination between 0.70 and 4.83. The estimated item difficulties for French are concentrated at lower ability levels, implying that the French test provides the most measurement precision for below- to average-ability students and less precision in the upper tail.

Overall, item discriminations were positive and of meaningful magnitude across all subjects, indicating that most items differentiated higher- and lower-performing students in the expected direction. Only a small fraction of items displayed weak discrimination or extreme difficulty values, suggesting generally sound psychometric quality.

[11] The idea with the year 2 evaluation was to follow students from year 1 with Grade 1 students advancing to Grade 2.

Table 4: *Item Map for Question Items by Exam and Grade Level*

	Year 1		Year 2	
	Baseline Exam Cohort 1	Endline Exam Cohort 1	Baseline Exam Cohort 2	Endline Exam Cohort 1 & 2
Arabic				
Grade 1	36 Q (19 A) [262 S]	35 Q (19 A) [241 S]	Untested	Untested
Grade 2	34 Q (9 A) [273 S]	33 Q (18 A) [254 S]	35 Q (9 A) [64 S]	33 Q (9 A) [231 S]
Grade 3	31 Q (14 A) [272 S]	34 Q (23 A) [262 S]	33 Q (13 A) [64 S]	34 Q (16 A) [237 S]
Grade 4	38 Q (15 A) [272 S]	34 Q (21 A) [263 S]	34 Q (16 A) [63 S]	34 Q (14 A) [235 S]
Grade 5	43 Q (6 A) [275 S]	35 Q (13 A) [266 S]	34 Q (14 A) [62 S]	35 Q (16 A) [234 S]
Grade 6	43 Q (17 A) [274 S]	38 Q (31 A) [263 S]	35 Q (17 A) [61 S]	37 Q (12 A) [235 S]
French				
Grade 1	63 Q (26 A) [254 S]	49 Q (18 A) [245 S]	Untested	Untested
Grade 2	72 Q (20 A) [260 S]^{A}	50 Q (32 A) [252 S]^{A}	49 Q (7 A) [66 S]	50 Q (10 A) [224 S]^{A}
Grade 3	72 Q (20 A) [270 S]^{A}	50 Q (32 A) [264 S]^{A}	50 Q (8 A) [65 S]^{C}	50 Q (10 A) [231 S]^{A}
Grade 4	45 Q (7 A) [271 S]^{B}	35 Q (20 A) [264 S]^{B}	50 Q (8 A) [65 S]^{C}	35 Q (10 A) [233 S]^{B}
Grade 5	45 Q (7 A) [268 S] ^{B}	35 Q (20 A) [258 S]^{B}	35 Q (9 A) [64 S]^{D}	35 Q (10 A) [235 S]^{B}
Grade 6	45 Q (7 A) [268 S]^{B}	35 Q (20 A) [260 S]^{B}	35 Q (9 A) [63 S]^{D}	35 Q (10 A) [231 S]^{B}
Mathematics				
Grade 1	38 Q (14 A) [240 S]	26 Q (12 A) [236 S]	Untested	Untested
Grade 2	38 Q (11 A) [266 S]	25 Q (16 A) [263 S]	26 Q (7 A) [67 S]	28 Q (8 A) [222 S]
Grade 3	48 Q (14 A) [266 S]	29 Q (18 A) [264 S]	27 Q (8 A) [63 S]	29 Q (12 A) [237 S]
Grade 4	61 Q (14 A) [270 S]	28 Q (18 A) [268 S]	27 Q (7 A) [67 S]	29 Q (13 A) [235 S]
Grade 5	66 Q (16 A) [269 S]	32 Q (22 A) [267 S]	28 Q (5 A) [65 S]	34 Q (12 A) [238 S]
Grade 6	59 Q (12 A) [269 S]	27 Q (22 A) [265 S]	32 Q (7 A) [64 S]	26 Q (8 A) [237 S]

Note: This table presents the number of question items for each exam administered at each grade level across the three subjects. The first number is number of question items (Q) on the exam. In parentheses are the number of anchor item questions (A) on the exam. The brackets includes the number of schools (S) that administered this grade level exam. *A*: Same test for Grade 2 and 3, *B*: Same test for Grade 4, 5 and 6. *C*: Same test for Grade 3 and 4. *D*: Same test for Grade 5 and 6.

C Differential Attrition

In this section, we report on the rates of differential attrition for each sample. Table 5 presents the rate of attrition from baseline survey to the endline survey sample of students across the various samples. Further, we show the differential attrition as the difference in attrition rate from the comparison schools compared to PSP school students. In each sample, the comparison group has a larger attrition rate. But this difference is not statistically significant nor are the differences considerably large. Attrition is highest amongst the Cohort 1 year 2 sample at roughly 15 percent of students, while 1 year sample attrition is much lower at roughly 5 percent for both Cohort 1 and Cohort 2. Differential attrition does not exceed 2.5 percent for any sample. This reassures us that the results of our analysis are not influenced by spurious relationships caused by varying levels of unobserved student performance for our outcome measures. In other words, the sample does not suffer from sample selection bias which may obfuscate the effect of the PSP.

Table 5: *Differential Attrition across Treatment Status*

	Control Attrition	Treatment Attrition	Differential Attrition
1-Year Exposure (Cohort 1)	.054	.041	.013
2-Year Exposure (Cohort 1)	.158	.145	.013
1-Year Exposure (Cohort 2)	.044	.021	.023
Pooled Exposure (Cohort 1 & 2)	.052	.038	.014

Note: This table reports the rate of attrition across the four samples for the comparison group, the treatment group, and the differential attrition rate of the comparison group minus the treatment group. None of these differences are statistically significant.

D Heterogeneity Analysis

We examine heterogeneous treatment effects of the ITT for the PSP program by the student's grade level. Table 6 presents the intent-to-treat effects sub-analysis by grade levels 2 through 6 across the four samples. Additionally, we perform heterogeneity analysis by sub-domains of learning within the subject exams for Arabic, French, and mathematics. Table 7 presents the ITT effect of the PSP on these sub-domains. Across all subjects, we explore the effect for exam items which are below the current grade level of the student, as well as exam items which are at the current grade level. For Arabic and French, we also examine how many words in each language the student correctly identified. And in mathematics, we explore student performance on several subsets of mathematical learning. Namely, calculation and numeracy; geometry, measures, and data; identification knowledge; and application and reasoning.

Table 6: *Intent-to-treat on student learning by grade and subject*

	1-Year Exposure (Cohort 1)				2-Year Exposure (Cohort 1)				1-Year Exposure (Cohort 2)				Pooled Exposure (Cohort 1 & 2)			
	Overall (1)	Arabic (2)	French (3)	Math (4)	Overall (5)	Arabic (6)	French (7)	Math (8)	Overall (9)	Arabic (10)	French (11)	Math (12)	Overall (13)	Arabic (14)	French (15)	Math (16)
Grade 2	0.818***	0.583***	1.196***	0.749***	0.554***	0.365**	0.869***	0.514***	0.491***	0.266	0.758***	0.495**	0.764***	0.533***	1.113***	0.700***
	(0.050)	(0.081)	(0.085)	(0.065)	(0.071)	(0.125)	(0.083)	(0.103)	(0.093)	(0.190)	(0.104)	(0.145)	(0.045)	(0.075)	(0.073)	(0.060)
	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.024]	[0.001]	[0.001]	[0.001]	[0.845]	[0.001]	[0.011]	[0.001]	[0.001]	[0.001]	[0.001]
Grade 3	0.794***	0.410***	1.011***	0.971***	0.586***	0.547***	0.819***	0.497***	0.382***	0.304	0.541***	0.380***	0.716***	0.386***	0.920***	0.856***
	(0.043)	(0.075)	(0.067)	(0.063)	(0.059)	(0.081)	(0.095)	(0.076)	(0.075)	(0.135)	(0.117)	(0.100)	(0.040)	(0.065)	(0.061)	(0.058)
	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.167]	[0.001]	[0.004]	[0.001]	[0.001]	[0.001]	[0.001]
Grade 4	1.011***	0.477***	1.410***	1.193***	0.599***	0.339***	0.807***	0.676***	0.360***	0.238	0.427***	0.439***	0.894***	0.432***	1.224***	1.054***
	(0.041)	(0.066)	(0.066)	(0.063)	(0.053)	(0.098)	(0.072)	(0.078)	(0.073)	(0.133)	(0.082)	(0.089)	(0.041)	(0.060)	(0.065)	(0.058)
	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.005]	[0.001]	[0.001]	[0.001]	[0.428]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]
Grade 5	0.873***	0.381***	1.356***	0.948***	0.681***	0.310***	1.047***	0.721***	0.512***	0.248	0.759***	0.572***	0.810***	0.362***	1.245***	0.878***
	(0.042)	(0.071)	(0.061)	(0.057)	(0.053)	(0.096)	(0.074)	(0.081)	(0.080)	(0.170)	(0.096)	(0.122)	(0.039)	(0.066)	(0.056)	(0.054)
	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.010]	[0.001]	[0.001]	[0.001]	[0.776]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]
Grade 6	0.918***	0.491***	1.419***	0.901***	0.549***	0.146	0.989***	0.543***	0.543***	0.333	0.773***	0.581***	0.851***	0.460***	1.295***	0.839***
	(0.039)	(0.068)	(0.057)	(0.058)	(0.049)	(0.088)	(0.064)	(0.074)	(0.086)	(0.150)	(0.105)	(0.132)	(0.037)	(0.062)	(0.055)	(0.054)
	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.511]	[0.001]	[0.001]	[0.001]	[0.182]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]
Number of Schools	276				184				62				338			

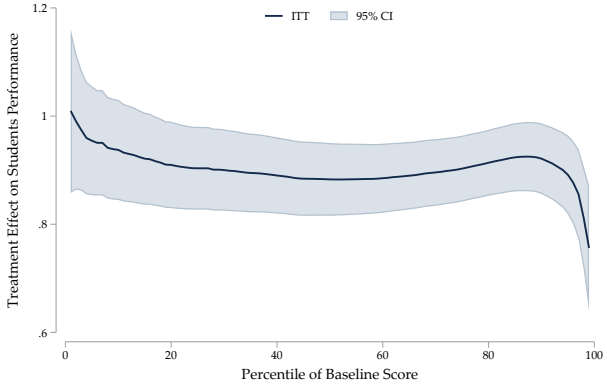
Note: This table presents the ITT effect of the PSP on IRT scores for each grades' sub-sample. The sample and outcome measures are similar in fashion to those presented in Table 1. Standard errors are clustered at the matched-pair level and shown in parentheses. P-values are computed via the wild cluster bootstrap-t (pairs as clusters) and are reported in brackets. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: *ITT Effects on Subdomains of Student Learning*

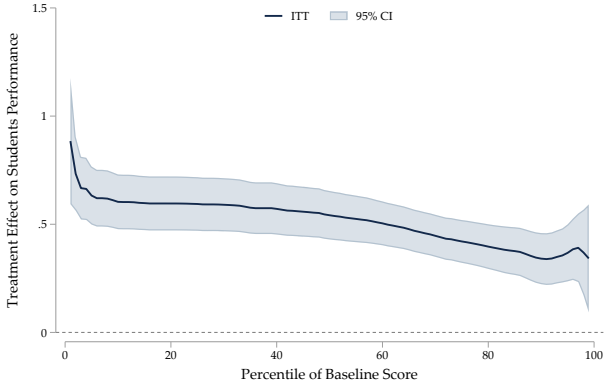
	2-Year Exposure (Cohort 1) (1)	1-Year Exposure (Cohort 2) (2)	Pooled Exposure (Cohorts 1 & 2) (3)
Panel A: Arabic			
Below-grade items	0.352***	0.260**	0.318***
	(0.082)	(0.116)	(0.067)
At-grade items	0.354***	0.231*	0.322***
	(0.070)	(0.133)	(0.062)
Number of words read correctly	6.295***	-1.055	5.792***
	(1.054)	(4.252)	(1.029)
Panel B: French			
Below-grade items	0.803***	0.597***	0.738***
	(0.085)	(0.081)	(0.071)
At-grade items	0.902***	0.648***	0.822***
	(0.087)	(0.085)	(0.074)
Number of words read correctly	8.733***	7.600***	8.401***
	(2.369)	(1.872)	(1.756)
Panel C: Mathematics			
Below-grade items	0.659***	0.573***	0.632***
	(0.065)	(0.077)	(0.050)
At-grade items	0.554***	0.386***	0.500***
	(0.060)	(0.079)	(0.049)
Calculation and numeracy	0.732***	0.625***	0.701***
	(0.069)	(0.082)	(0.054)
Geometry, measures, data	0.385***	0.328***	0.369***
	(0.061)	(0.090)	(0.051)
Knowing	0.843***	0.545***	0.719***
	(0.092)	(0.092)	(0.068)
Applying and reasoning	0.515***	0.485***	0.507***
	(0.061)	(0.080)	(0.049)
Number of Students	11152	4053	15205
Number of Schools	186	62	246

Note: This table presents the ITT effect of the PSP on sub-domains of item questions for each of the three subject exams: Arabic, French, and math. All of the outcome measures are measured as an IRT score except for the number of words read correctly which is a count variable. Standard errors are clustered at the matched-pair level and shown in parentheses. P-values are computed via the wild cluster bootstrap-t (pairs as clusters) and are reported in brackets. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

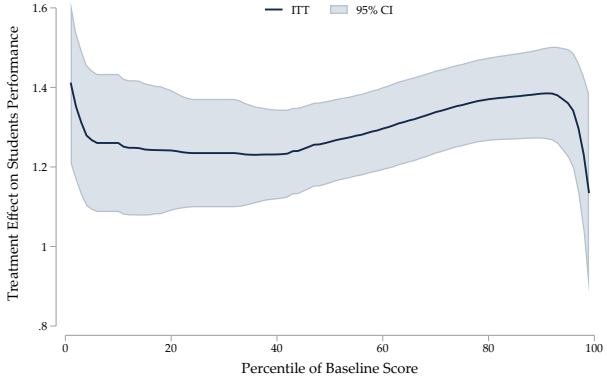
Figure 9: *Cohort 1: ITT Effects by Baseline Achievement Percentile After One Year of PSP Exposure*



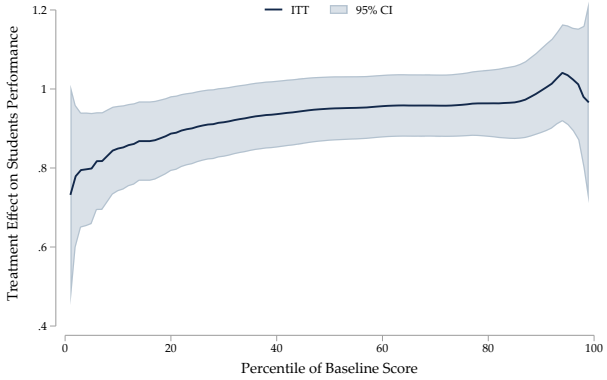
Overall



Arabic



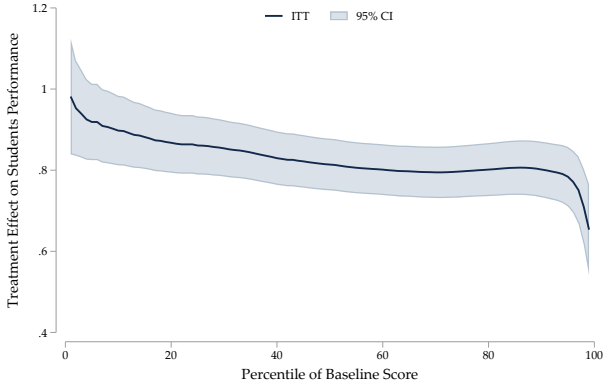
French



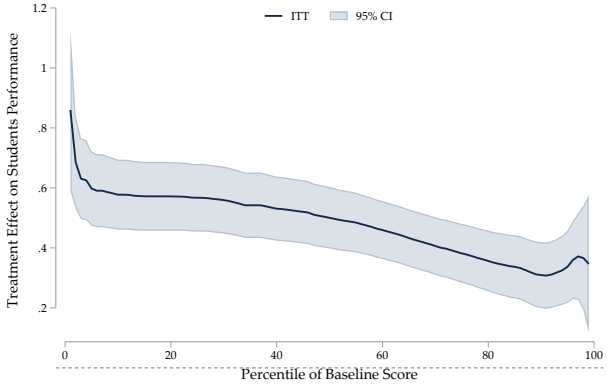
Math

Note: This figure shows the ITT effect at each Quantile of the baseline test score distribution. This is a form of local rolling regression. The black line is the ITT effect and the shaded gray area are the 95 percent confidence interval for the estimated effect. This figure only estimates the effects for the Cohort 1 sample after one year of exposure to the PSP.

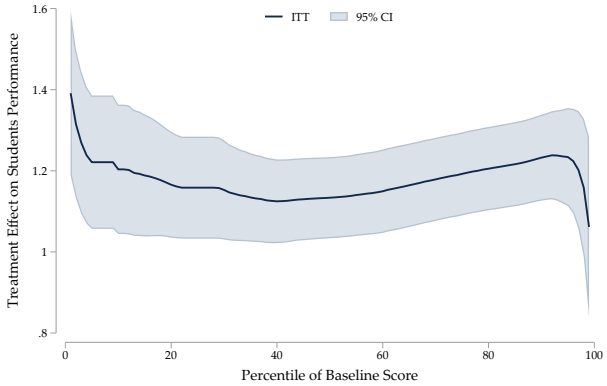
Figure 10: *Pooled Sample: ITT Effects by Baseline Achievement Percentile After One Year of PSP Exposure*



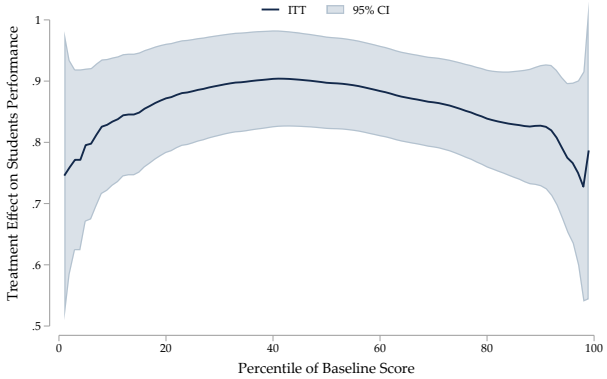
Overall



Arabic



French



Math

Note: This figure shows the ITT effect at each Quantile of the baseline test score distribution. This is a form of local rolling regression. The black line is the ITT effect and the shaded gray area are the 95 percent confidence interval for the estimated effect. This figure only estimates the effects for the pooled sample of both Cohort 1 and 2 after one year of exposure to the PSP.

E Event Study

Since the ceiling effects are present over the two years, it is important to consider the time dynamics of treatment, which may alter the impact of the PSP. To address this, we present an event study estimation modifying our preferred specification in Equation (1). We also do this separately for "at grade" and "below grade" items (questions) on the exams in order to look for patterns in the impact of the PSP on student learning. We estimate the specification in Equation (2):

$$Y_{igsjt} = \beta_1(Treat_{sj} \times 1[Year_t = 1]) + \beta_2(Treat_{sj} \times 1[Year_t = 2]) + Year_t + \delta X'_{igsj} + \gamma_{igsj} + \varepsilon_{igsjt} \quad (2)$$

Y_{igsjt} is the test score performance for student i in grade g in school s in matched pair j in period t. $Treat_{sj}$ is the assignment of school s to receive the education intervention. $Year_t$ is a categorical indicator of the year the student's test is conducted. This captures the change in test performance for all students over the two years of the study, and so this shows how the performance changed over time for the students in the comparison schools. The treatment effects of the PSP program in the first and second year of exposure are measured by β_1 and β_2, respectively. The γ_{igsj} term is a student fixed effect, which measures the impacts on Y_{igsjt} of all student characteristics, observed or unobserved, that do not change over time. However, it is possible that the influences of some student characteristics change over time, so a vector of observed student characteristics, denoted by X_{igsj}, estimate the impacts of these variables. The corresponding vector of coefficients for this interaction term, denoted by δ, measures the contributions of observed student characteristics to changes in the outcome variables over time. Since nearly all students stay in the same school between baseline and endline data collection, student fixed effects also include school fixed effects, and so the X_{igsj} variables can also include observed school variables. The student variables available in the dataset are students' gender and whether the student was eligible for a social safety net measure (measured at baseline). The school characteristics are school-level average test scores, region, average class size, and the student-to-teacher ratio (all measured at baseline). A LASSO is used to determine in a data-driven way with X_{igsj} interactions to include in the specification.[12]

Table 8 presents estimates of Equation (2) for the regular IRT scores (columns (1) – (4)), for IRT scores constructed using only items (questions) that are at grade level (columns (5) – (8), and for IRT scores constructed using only items that are below grade level (columns (9) – (12)). Column (1) shows, for regular IRT scores, that the treatment effect for the first year is 0.92 standard deviations (s.d.). The second year treatment effect is smaller: 0.55 s.d.. These estimated effects are very similar in size to those in Table 1; the slight differences are due to the constraint in Table 8 that the impacts of all other variables are the same for years 1 and 2, while the estimates in Table 1 are separate for each year and so do not impose this constraint. The coefficients for Year 1 and Year 2 in column (1) of Table 8 show the

[12]More specifically, we first calculate each student's change in test scores between the baseline and endline assessments. Then, we residualize this change in test scores, subtracting the comparison group's grade-by-matched pair mean. Lastly, with these residuals, we use a post-double-selection LASSO to select relevant predictors X_{igsj} (Belloni et al., 2011).

improvement in test scores for the comparison group students; compared to the baseline, their scores are 0.17 s.d. higher in year 1 and 0.22 s.d. higher in year 2, relative to their baseline test scores. This confirms the visual results in Figure 5 that comparison school students are increasing their learning over time, but the 0.17 s.d. increase from baseline to endline in year 1 is higher than the 0.05 s.d. increase (0.22 – 0.17) from endline in year 1 to endline in year 2 (which is also seen in the dashed lines in Figure 5 – the move to the right from baseline to endline of year 1 is larger than the change from endline of year 1 to endline of year 2). Table 8 also shows, as one would expect, that students in higher grades have higher test scores than students in lower grades.

The remaining columns in Table 8 investigate whether there are different effects of the PSP reform for items (questions) that are at grade level, compared to items that are below grade level. If the PSP intervention helps only students who have fallen behind, we would expect larger effects on below grade level items. Columns 5 – 8 estimate Equation (2) for student test performance on items that are at grade level. The impact of PSP on at grade level performance is very similar to the overall results. For overall scores, year 1 PSP exposure has a 0.73 s.d. treatment effect while year 2 PSP exposure only has a 0.41 s.d. effect. And the time effects are quite similar with year 1 having 0.41 s.d. higher test scores and year 2 having 0.32 s.d. higher test scores relative to comparison school baseline IRT test score of -1.02. This contrasts the below grade level student test performance in columns 9 – 12. While treatment effects are very similar for below-grade level items and at-grade level items, the year effects are much smaller. For overall IRT test scores, year 1 PSP exposure has a 0.759 s.d. effect and year 2 PSP exposure has a 0.485 effect relative to comparison baseline IRT test score average of -0.8 s.d.. The improvement in overall test scores over time relative to baseline test scores is only 0.395 s.d. in the first year and 0.450 s.d. in the second year. These time effects are nearly half the size of the regular IRT score and at grade level IRT score estimated time effects. This suggests that students' improved performance on items below grade level is not quite as large as on items at the student's grade level. But the effect of the PSP is still improving test performance for items at grade level and below grade level to a similar degree over time meaning that PSP is benefiting both low-performing and high-performing students.

In general, the estimated impacts for test items that are at grade level in columns (5) – (8) are similar to those for items below grade level in columns (9) – (12). For example, the overall impacts of the PSP after one and two years are 0.72 and 0.42, respectively, for at-grade items, and 0.76 and 0.49, respectively, for below-grade items, which are quite similar. However, there are a few differences that are interesting. First, the overall year 2 coefficient for at-grade items, 0.32, is noticeably smaller than the overall year 2 coefficient for below-grade items, 0.45. This suggests that learning among the comparison school students in year 2 was stronger on below-grade items than on at-grade items, so the focus in year 2 may have been more on below-grade items. A closer examination reveals that this difference is coming primarily from French and mathematics; the impact on Arabic is much more in the opposite direction, which could reflect that students are already close to at grade for Arabic. Another interesting difference is that, for mathematics in year 2, the effect of the PSP is much stronger for below grade items (0.57) than for at grade items (0.40); perhaps the PSP switched to emphasizing more below-grade instruction in mathematics in the second year, which may also explain why the overall PSP impact on mathematics is lower in year

2; if many students were brought up to "at grade" level in year 1, an emphasis on "below grade level" in year 2 would have a smaller effect on the students who were at grade level in year 2.

In summary, this examination of at-grade and below-grade items (questions) on the exams does not provide any clear explanation for the decline in the impacts of the PSP from year 1 to year 2. Our event study analysis confirms that time dynamics are at play in partially explaining the change in ITT effects from year 1 to year 2. Further analysis is needed to understand this reduction in the impact of the PSP over time.

Table 8: *Event Study Design for ITT Effect of PSP Program on Students Performance (IRT Scores)*

	2-Year Exposure (Cohort 1)				2-Year Exposure (Cohort 1): At Grade				2-Year Exposure (Cohort 1): Below Grade			
	Overall (1)	Arabic (2)	French (3)	Math (4)	Overall (5)	Arabic (6)	French (7)	Math (8)	Overall (9)	Arabic (10)	French (11)	Math (12)
Year 1 x Treatment	0.924***	0.574***	1.269***	0.936***	0.724***	0.456***	0.927***	0.798***	0.759***	0.415***	1.020***	0.782***
	(0.043)	(0.059)	(0.068)	(0.053)	(0.040)	(0.060)	(0.063)	(0.044)	(0.033)	(0.058)	(0.052)	(0.040)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Year 2 x Treatment	0.550***	0.320***	0.833***	0.572***	0.421***	0.299***	0.632***	0.403***	0.485***	0.297***	0.573***	0.565***
	(0.051)	(0.067)	(0.062)	(0.070)	(0.042)	(0.057)	(0.051)	(0.055)	(0.040)	(0.063)	(0.051)	(0.062)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Year 1	0.174***	-0.017	0.210***	0.376***	0.414***	0.513***	0.198***	0.590***	0.395***	0.221***	0.229***	0.678***
	(0.039)	(0.045)	(0.051)	(0.050)	(0.032)	(0.042)	(0.044)	(0.039)	(0.029)	(0.044)	(0.038)	(0.038)
	[0.000]	[0.703]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Year 2	0.224***	-0.115	0.353***	0.635***	0.322***	0.497***	0.229**	0.733***	0.450***	0.160*	0.458***	0.848***
	(0.081)	(0.085)	(0.112)	(0.125)	(0.070)	(0.069)	(0.098)	(0.099)	(0.063)	(0.089)	(0.096)	(0.109)
	[0.007]	[0.179]	[0.002]	[0.000]	[0.000]	[0.000]	[0.022]	[0.000]	[0.000]	[0.077]	[0.000]	[0.000]
Grade 2	0.669***	0.722***	0.335***	-0.043								
	(0.106)	(0.081)	(0.098)	(0.123)								
	[0.000]	[0.000]	[0.001]	[0.726]								
Grade 3	1.006***	0.400***	1.305***	-0.121	0.445***	0.107*	0.961***	-0.380***	0.207***	-0.510***	0.557***	0.338***
	(0.142)	(0.152)	(0.190)	(0.242)	(0.060)	(0.059)	(0.085)	(0.091)	(0.053)	(0.077)	(0.076)	(0.108)
	[0.000]	[0.010]	[0.000]	[0.618]	[0.000]	[0.071]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.002]
Grade 4	1.682***	0.434*	2.079***	0.481	1.008***	-0.080	1.753***	0.159	0.481***	-0.426***	0.472***	0.774***
	(0.198)	(0.224)	(0.283)	(0.360)	(0.121)	(0.114)	(0.171)	(0.182)	(0.105)	(0.153)	(0.153)	(0.216)
	[0.000]	[0.056]	[0.000]	[0.185]	[0.000]	[0.489]	[0.000]	[0.385]	[0.000]	[0.007]	[0.003]	[0.001]
Grade 5	2.010***	0.462	2.378***	0.604	1.201***	-0.421**	1.778***	0.445	0.464***	-0.472**	0.551**	0.488
	(0.261)	(0.298)	(0.380)	(0.479)	(0.184)	(0.172)	(0.261)	(0.273)	(0.158)	(0.231)	(0.232)	(0.324)
	[0.000]	[0.124]	[0.000]	[0.210]	[0.000]	[0.016]	[0.000]	[0.106]	[0.004]	[0.044]	[0.020]	[0.136]
Grade 6	2.743***	1.703***	2.715***	0.840	1.933***	0.289	2.315***	1.125***	1.302***	0.815***	0.565	0.962**
	(0.311)	(0.369)	(0.514)	(0.558)	(0.229)	(0.228)	(0.332)	(0.346)	(0.308)	(0.307)	(0.354)	(0.432)
	[0.000]	[0.000]	[0.000]	[0.136]	[0.000]	[0.208]	[0.000]	[0.002]	[0.000]	[0.009]	[0.114]	[0.029]
Number of Students	11,150	3,726	3,695	3,729	11,150	3,726	3,695	3,729	11,150	3,726	3,695	3,729
Baseline Mean (Control)	-0.670	-0.450	-0.680	-0.870	-0.990	-0.950	-0.860	-1.160	-0.800	-0.560	-0.840	-0.930
Baseline SD (Control)	1.010	0.940	1.060	0.990	0.690	0.700	0.710	0.620	0.420	0.380	0.430	0.380
Year 1 Mean (Control)	-0.100	-0.120	-0.110	-0.070	-0.430	-0.280	-0.550	-0.460	-0.420	-0.270	-0.600	-0.400
Year 1 SD (Control)	0.990	0.970	1.010	0.990	0.840	0.870	0.820	0.810	0.790	0.820	0.740	0.780
Year 2 Mean (Control)	0.140	0.110	0.190	0.120	-0.320	-0.180	-0.440	-0.350	-0.290	-0.090	-0.430	-0.350
Year 2 SD (Control)	0.970	1.000	0.930	0.980	0.790	0.860	0.730	0.740	0.840	0.890	0.750	0.840

Note: This table reports on the program's intent-to-treat (ITT) effects in Cohort 1 who were tracked over two years of exposure to the PSP program. The first four columns estimate the IRT score for student performance. Columns 5 - 8 estimates the effect for IRT scores considering the students' performance on question items that are at their current grade level, while Columns 9 - 12 consider the students' performance on question items with difficulty below their current grade level. Each subject level exams (Arabic, French, and Math) is reported alongside a composite score stacking the three sub-samples. Standard errors are clustered at the matched-pair level and shown in parentheses. Adjusted p-values using the Westfall–Young step-down procedure are reported in brackets. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

F Anchor Item Analysis

Item response to anchor items. A potential alternative explanation for the change in the ITT effects from year 1 to year 2 may come from the composition of the assessment (test) items. For example, if we observe lower ITT effects in year 2 compared to year 1, it may be random chance that the PSP students performed worse on the subset of items (questions) in the year 2 exam than on the items in the year 1 exam, or vice versa. This change in the composition of the test may not accurately reflect the true ability of the students. To investigate this possibility, we estimate the ITT effects for the anchor items. These are the items (questions) that are asked of the students on all three exams (baseline, year 1, and year 2). By using the same items for all three years, the composition of the test questions does not change over time and is unaffected by this concern. Unfortunately, limiting the analysis to the anchor items alone prevents us from constructing IRT scores for this subset of questions. The IRT questions require a log-likelihood search process to parametrically fit our two parameter IRT model. With a limited set of items, the model was unable to converge and we cannot fit the model. Instead, we present the ITT effect on anchor items in Table 9 using summary score percentages (e.g., the percentage of correctly answered questions) standardized relative to the comparison school students' endline performance. This can be interpreted in the same way as the IRT estimates, that is in terms of a standard deviation change in student skills. Table 9 shows that, in general, the item composition of the exams is not driving the change in ITT effects from year 1 to year 2. Consistent with the IRT ITT effects in Table 1, year 1 ITT effects are of similar magnitude and are larger than the year 2 ITT effects. In year 1, averaging across all subjects, PSP students performed 0.64 s.d. better than comparison students, and 0.47 s.d. better in year 2. However, this does not hold for all subjects; the impact for mathematics is slightly larger in year 2 than in year 1. This rules out that the characteristics of the exam changing over time explain the change in results.

Table 9: *ITT Effect of PSP Program on Students Performance (Anchor Item Summary Scores)*

	1-Year Exposure (Cohort 1)				2-Year Exposure (Cohort 1)			
	Overall (1)	Arabic (2)	French (3)	Math (4)	Overall (5)	Arabic (6)	French (7)	Math (8)
Panel A: Overall								
All students	0.635***	0.311***	1.258***	0.577***	0.465***	0.238***	0.602***	0.621***
	(0.035)	(0.046)	(0.087)	(0.044)	(0.038)	(0.059)	(0.056)	(0.054)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Number of Students	11,150	3,726	3,695	3,729	11,150	3,726	3,695	3,729
Baseline Mean (Control)	-0.520	-0.530	-0.540	-0.490	-0.560	-0.520	-0.790	-0.460

Note: This table reports on the program's intent-to-treat (ITT) effects in Cohort 1 who were tracked over two years of exposure to the PSP program. The columns estimate the summary test score for student performance only on anchor item questions. Each subject level exams (Arabic, French, and Math) is reported alongside a composite score stacking the three sub-samples. The outcome measure is a summary score (e.g., percentage correctly answered) relative to the comparison mean. Therefore it can be interpreted as a standard deviation change relative to the comparison group. Standard errors are clustered at the matched-pair level and shown in parentheses. Adjusted p-values using the Westfall–Young step-down procedure are reported in brackets.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

G Untestable Hypotheses

Alongside our mechanism analysis, we discuss several hypotheses that cannot be empirically tested with the available data.

Implementation intensity and teacher incentives. The implementation of the PSP may have differed between the first and second years as the program scaled from a relatively small rollout (626 schools) to a much larger national expansion (2,626 schools). We identify four dimensions along which implementation differed across these two years. While we cannot test the magnitude of these differences with the available data, access to school or classroom-level implementation data would allow us to assess whether these dimensions changed meaningfully from Year 1 to Year 2.

1. **Teacher Training Quality and Intensity.** According to implementation teams, teacher training and support may have been more 'intensive' in Year 1, when the implementation sample was smaller. This may be reflected in both the selection of trainers (inspectors) – who were among the most qualified and motivated within the inspectorate – and the duration and structure of their training. The Year 1 training cycle was longer and included a hands-on practicum with students, allowing inspectors – who are typically not in direct contact with classrooms on a daily basis – to practice and internalize the pedagogical techniques before supporting teachers. In contrast, the Year 2 training was shorter and did not include a practicum component, potentially reducing inspectors' capacity to support teachers.

2. **Frequency and Quality of School Support Visits.** According to implementation teams, inspectors conducted more school visits in Year 1 than in Year 2. As a result, the intensity and quality of teacher support were higher in the first year, providing more consistent feedback and reinforcement of program practices.

3. **Incentive Structure for Participating Schools.** The PSP included an annual financial incentive for the schools implementing the program. This financial incentive was awarded to schools in Year 1. However, this incentive applied only to Year 1, which may have affected teachers' motivation and engagement in the second year.

   On the other hand, in Year 1, a national teacher strike occurred, to which PSP teachers might have been less likely to participate in, if they were satisfied with the financial incentive their school received that year. However, initial evidence collected by the implementing teams suggested that strike participation rates were similar in both PSP and comparison schools.

4. **Availability of Instructional Materials.** One improvement in Year 2 compared to Year 1 was that textbooks and teaching materials were available from the start of the school year. In Year 1, these materials arrived later, which may have affected early implementation fidelity and limited students' access to textbooks at home during the initial months of the program. This factor would run counter to our results, as earlier availability of instructional materials would be expected to strengthen learning gains

in Year 2. While this pattern is not observed in the data, it is possible that any benefits from the improved availability of learning materials were offset by other factors in the second year.

Spillovers to comparison schools. Another possibility is that comparison schools experienced direct or indirect exposure to elements of the PSP between Years 1 and 2, for example through teacher mobility or informal diffusion of pedagogical practices. Such spillovers could, in principle, attenuate estimated treatment effects by narrowing differences between PSP and comparison schools. However, we do not have data on teacher transfers or classroom practices and therefore cannot assess the presence or magnitude of such spillovers. Moreover, given the multi-dimensional nature of the PSP – which combines structured pedagogy, targeted remediation, teacher training and specialization, and material support – it is unclear to what extent partial or indirect exposure to the program alone could explain the apparent "catch up" of the comparison schools to the PSP schools in year 2.